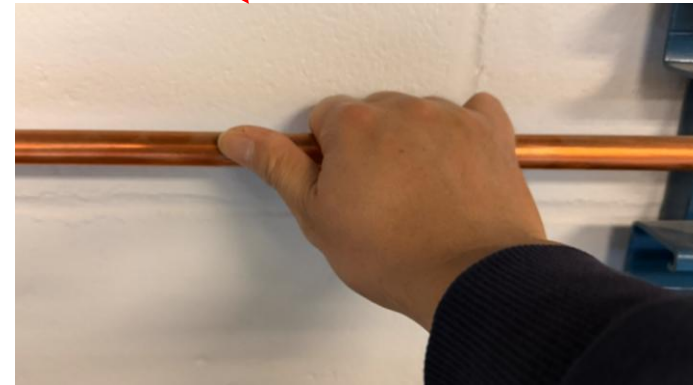
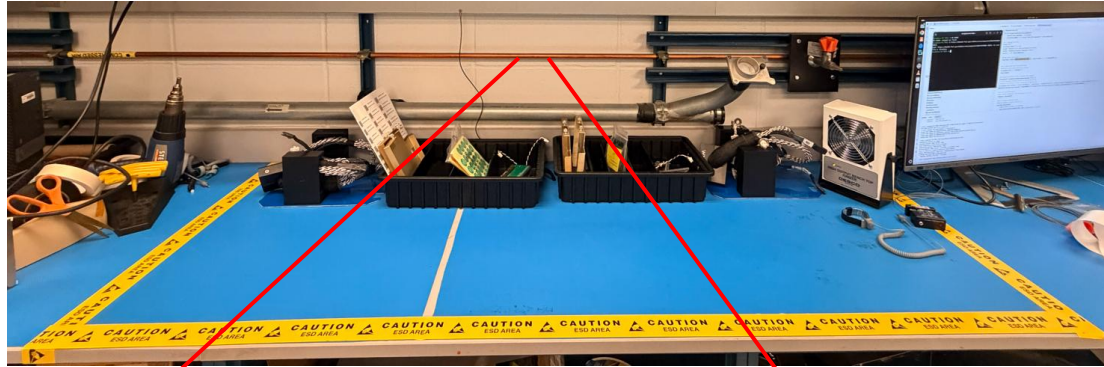


Pop-ups for FEMB QC Procedure

Karla Flores, Lingyun Ke



Discharge Yourself!



Both hands hold the bronze pipe firmly and softly for 3 seconds

(When you are **NOT** wearing on the wrist strap or ESG gloves, you **MUST** discharge yourself before you touch anything in the assembly area)

Initial Checkout

Please contact Tech Coordinator if any accessory is missing!



Safety First!



Long
Sleeve Top

Long Pants

Closed Toe
Shoes

ODH Monitor



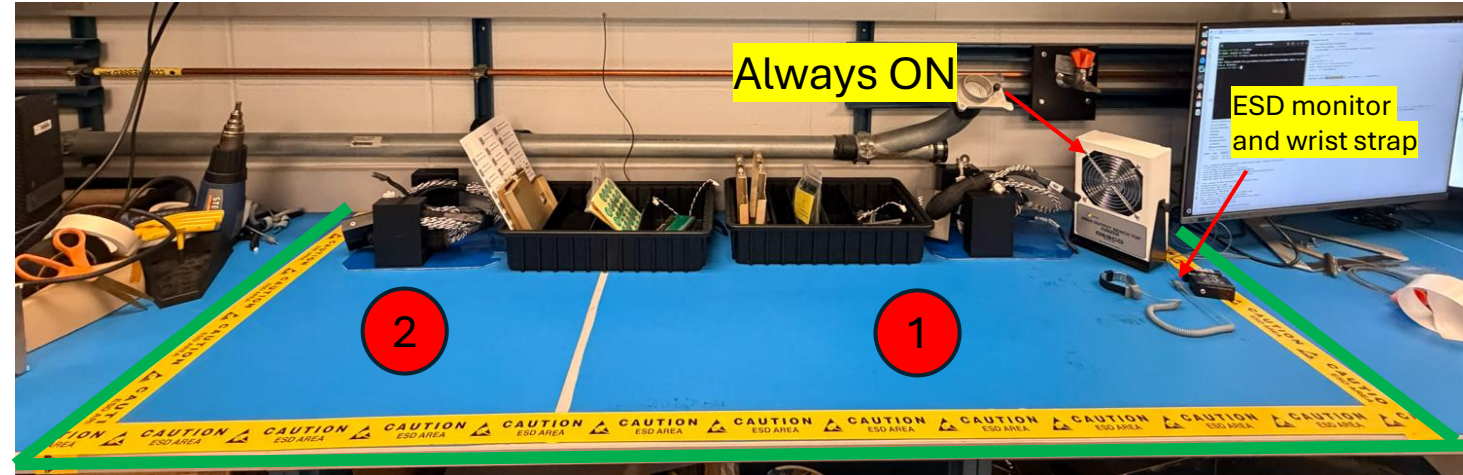
ESD-safe Gloves

Cryogenic Gloves



Anti-static
Wrist Strap

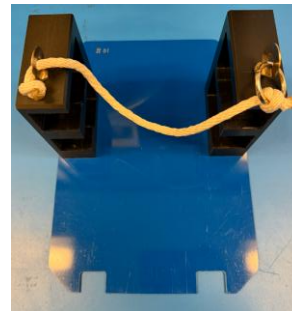
Cryogenic
Face Mask



1: Assembly area

2: Place the assembled CE box support structure (to be tested in the next run)

Is the assembly area neat and clean?



2 CE box
support structures



Cold cable bundle
(2 sets)



2 accessory trays

Checklist for accessory tray # 1

Please contact Tech Coordinator if any accessory is missing!

Item	Quantity
Screwdriver (PHILLIPS)	1
Allen wrench(3/16)	1
miniSAS cable clamp with 2 thumb nuts	1
Power cable clamp with 2 thumb nuts	1
Yellow Nylon Probe Pick Spudger	1
miniSAS cable clamp (VD)	2
M3x12 screws (VD)	>4
Thumb nut for miniSAS cable adapter (HD)	>4
miniSAS cable adapter	2
ToyTPC with cable attached	4
CE box fastener (2 unit/set)	2



screwdriver



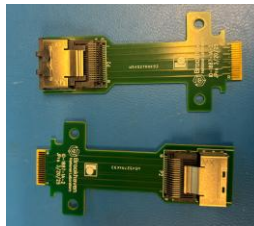
miniSAS cable clamp



power cable clamp



Spudger



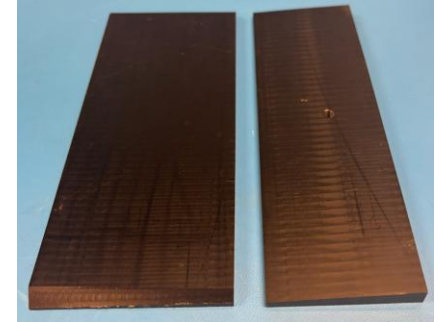
HD miniSAS
cable adapter



3/16 Allen wrench



CE box fastener (B)



CE box fastener (T)



Toy TPCs (B)



Toy TPCs (T)



VD M3x12

HD M3
thumb nuts

VD miniSAS
cable clamp

Checklist for accessory tray # 2

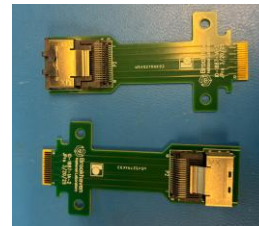
Please contact Tech Coordinator if any accessory is missing!

Item	Quantity
Screwdriver (PHILLIPS)	0
Allen wrench(3/16)	0
miniSAS cable clamp with 2 thumb nuts	0
Power cable clamp with 2 thumb nuts	0
Yellow Nylon Probe Pick Spudger	0
miniSAS cable clamp (VD)	2
M3x12 screws (VD)	>4
Thumb nut for miniSAS cable adapter (HD)	>4
miniSAS cable adapter	2
ToyTPC with cable attached	4
CE box fastener (2 unit/set)	2



In Tray#1

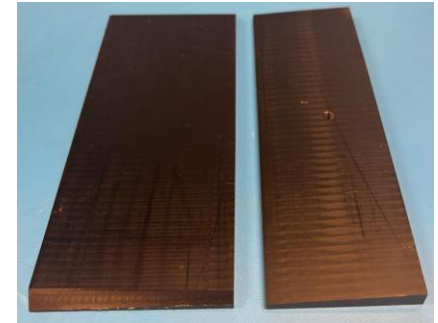
Check if teeth washers hold the thumb screws?



HD miniSAS cable adapter



CE box fastener (B)



CE box fastener (T)



Toy TPCs (B)



Toy TPCs (T)



VD M3x12

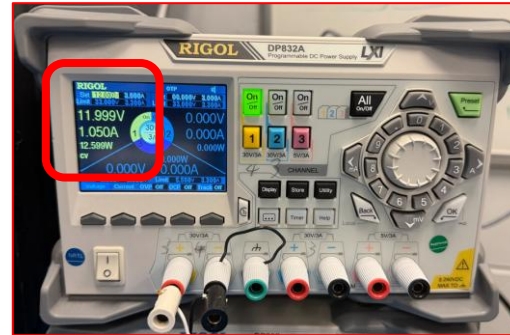
HD M3 thumb nuts

VD miniSAS cable clamp

CTS Initial Checkout

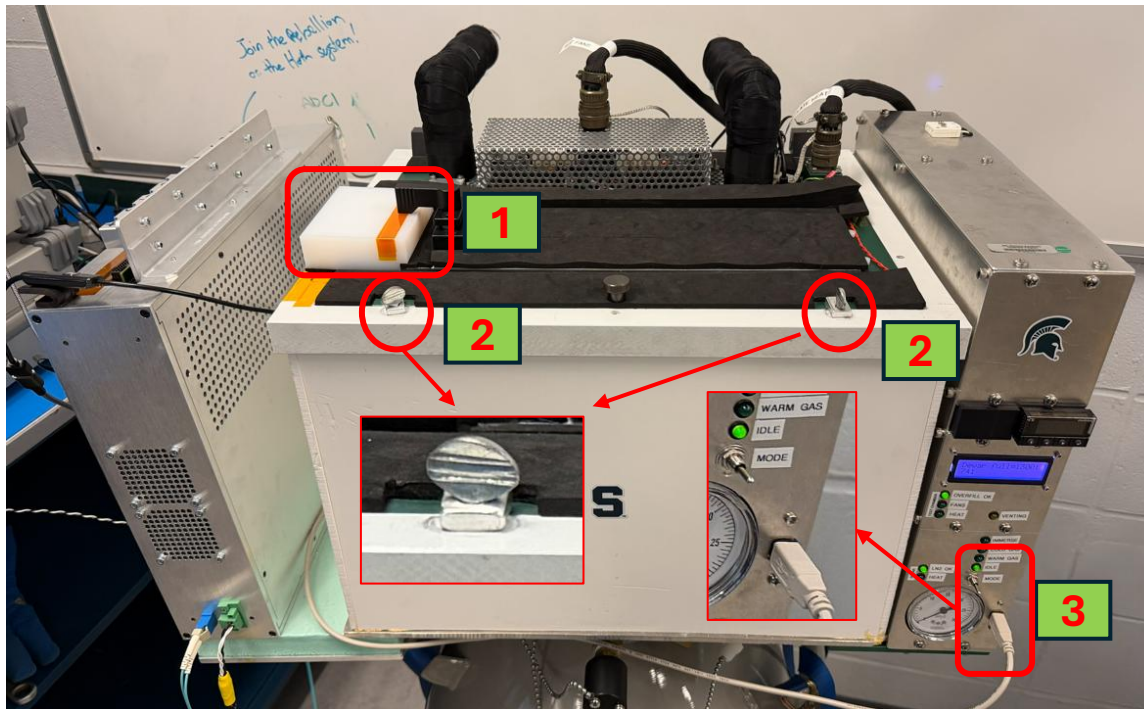
12V for CTS is ON.

Is current between 0.5A and 2A?

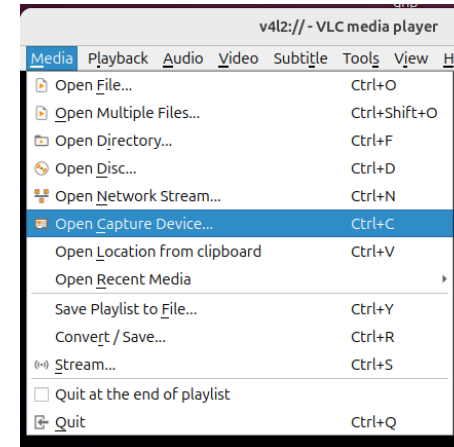


Is CTS chamber top close?

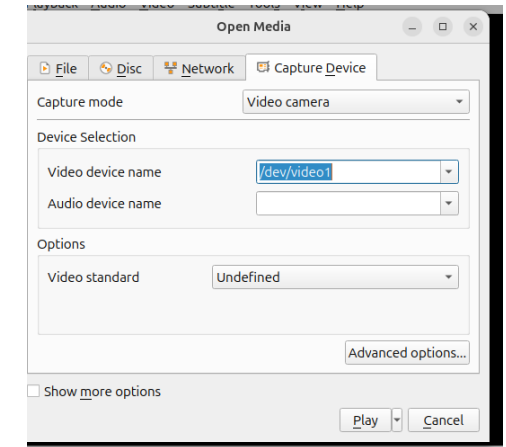
1. Is white block in position?
2. Is top locked?
3. LED 'IDLE' ON and USB connected?



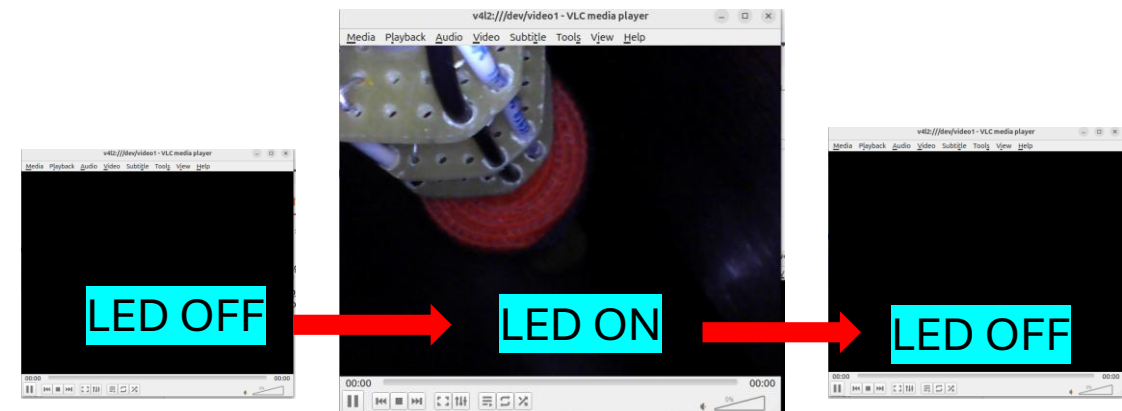
Is camera alive? (VLC player)
Turn on LED, check video and turn it off.



App VLC player
Open Capture Device...



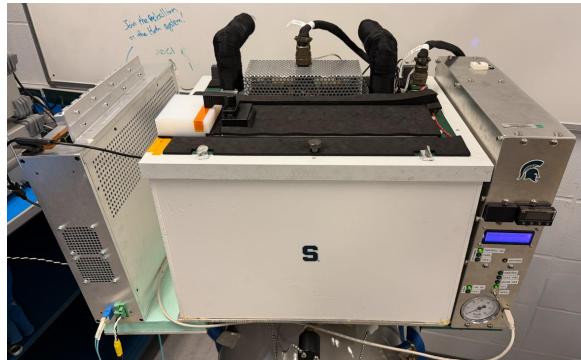
Select video device



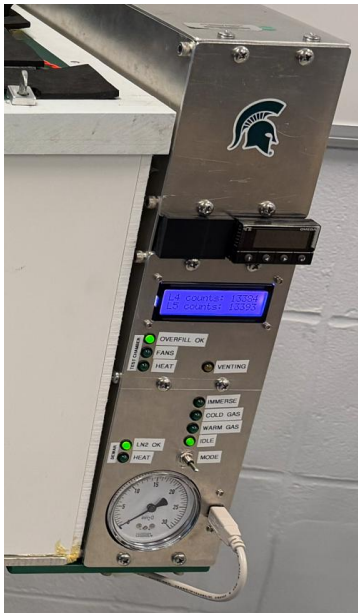
Computer to identify CTS device
Check LN2 level

LN2 Filling (**STOP** if you are **NOT** trained)

Follow these steps to fill the CTS dewar with LN2:

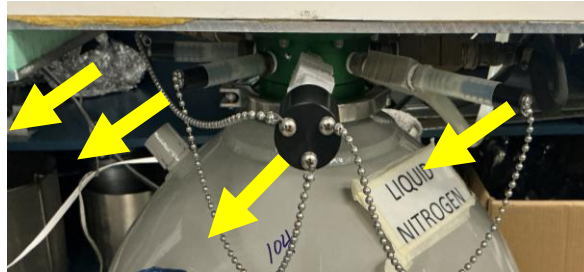


The top **MUST** be sealed!

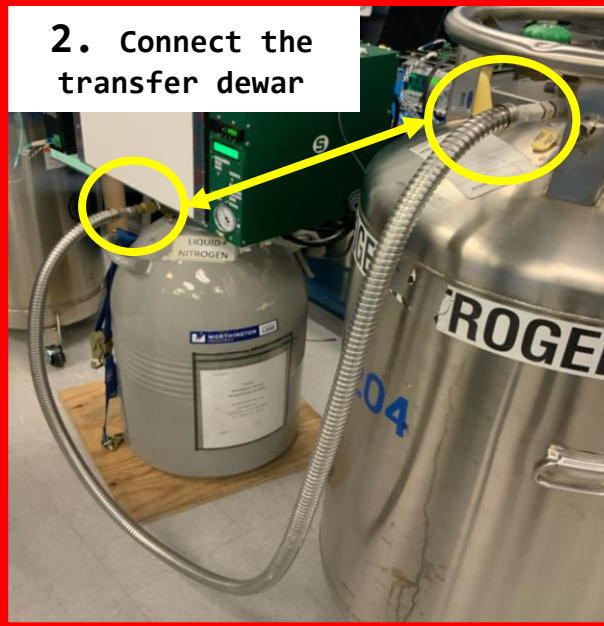


Power On!

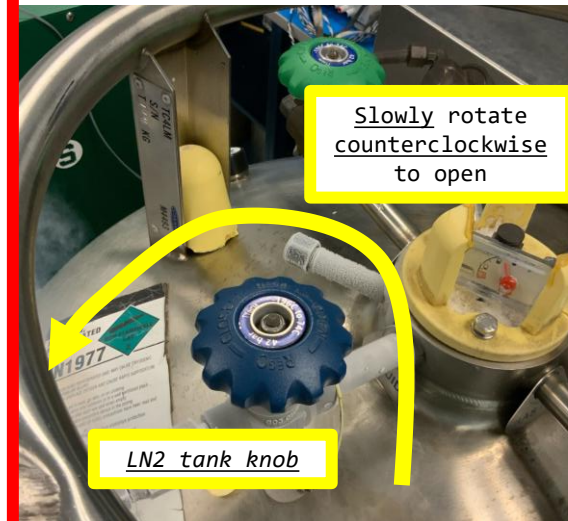
1. Remove the LN2 Dewar plugs
Keep CTS cover close



2. Connect the transfer dewar



3. Start the filling process

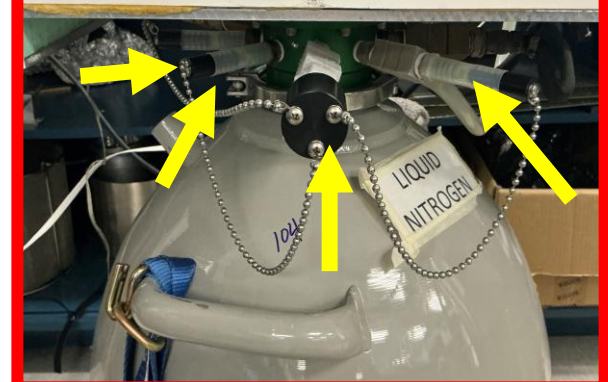


4. Monitor until "Dewar Full"
> 1800



5. Stop the filling process
(rotate LN2 tank knob clockwise)

5. Remove the transfer line
and place LN2 Dewar plugs
back.

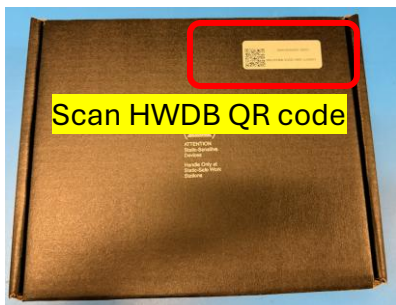


6. (1) CTS controlled by PC
Confirmed in the terminal. The chamber will be in warm mode for 20 minutes and switch to IDLE mode automatically.
(2) Manual operation
Set chamber in warm mode for 20 minutes and then switch to IDLE mode

Note: you may continue CE box assembly during this warm-up period.

CE Box Visual Inspection

Please contact Tech Coordinator if abnormality is observed!



Scan HWDB QR code



Is foam box broken?

Is CE box enclosed in the ESD bag?

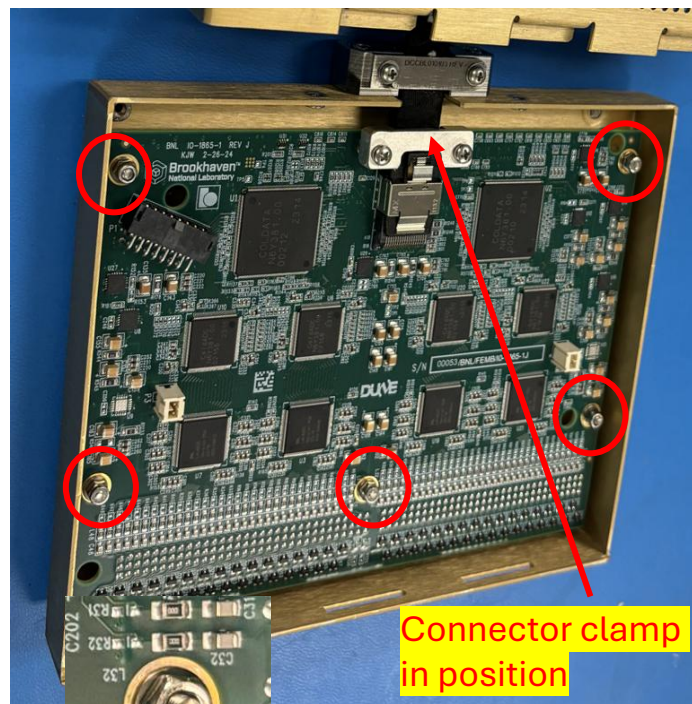


Scan CE box QR code (or entry CE box SN number)
(e.g. HD-10006)



Entry the last 4 digits of SN on CE box cover
(e.g. 0006 in the picture)

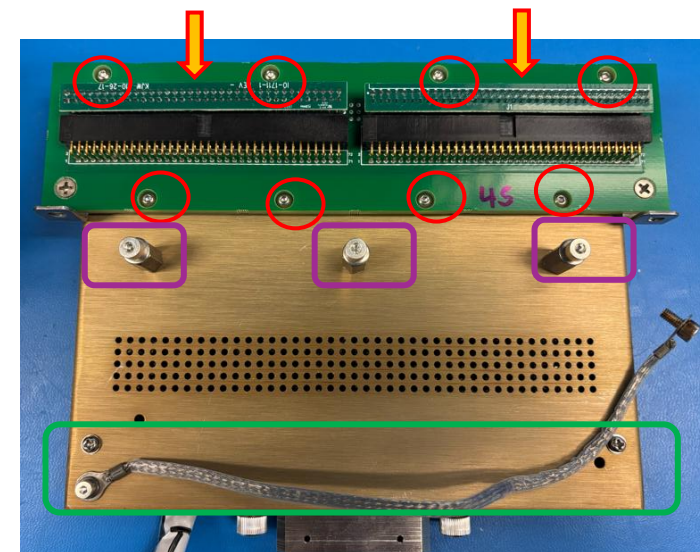
VD CE box (1865)



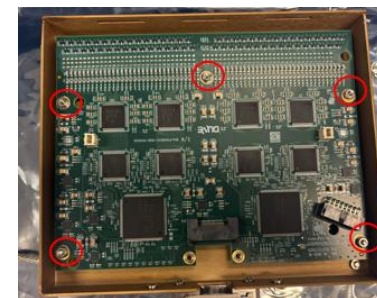
Connector clamp
in position

5 nuts fastened with
washers?

HD CE box (1826)



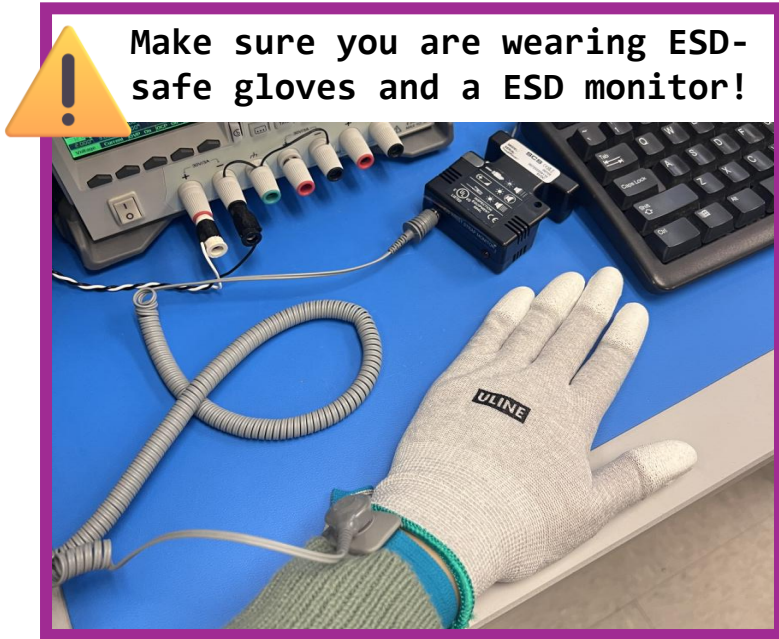
6 Nuts for CR-RC adapter board?
3 sets of lock nut and washer?
1 Ground strip?
2 FE input terminators?



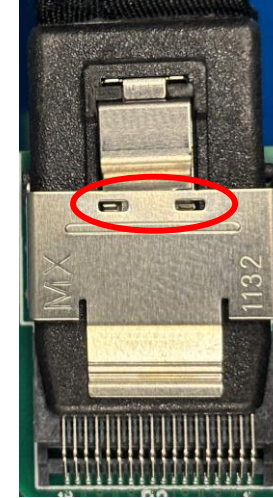
5 nuts fastened with
washers?

Bottom CE Box (VD) Assembly

Follow these steps to assemble the Bottom CE Box in the support structure:



1. Install miniSAS Cable + Clamp

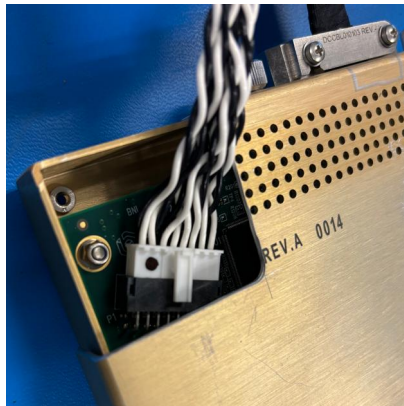


Locks in position

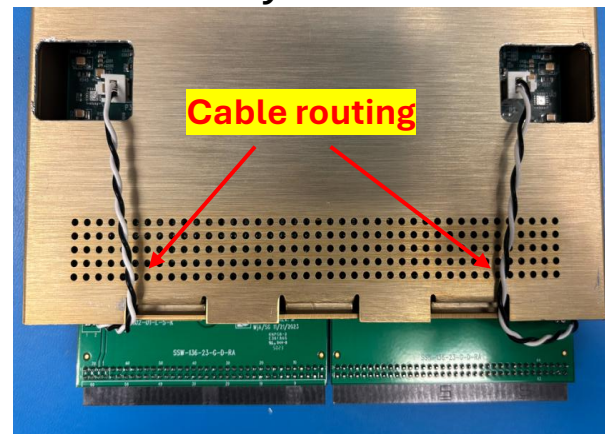
2. Install Test Cover



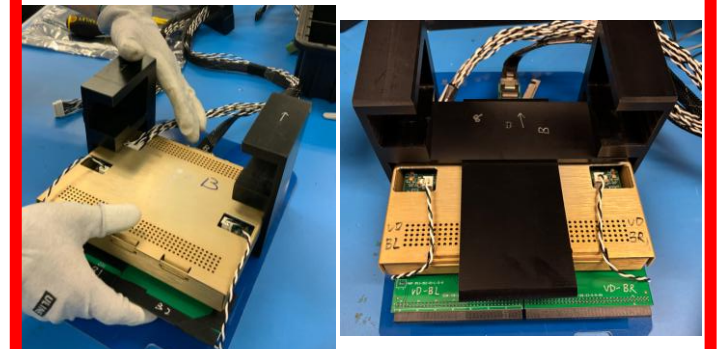
3. Install power cable



4. Install Toy TPCs

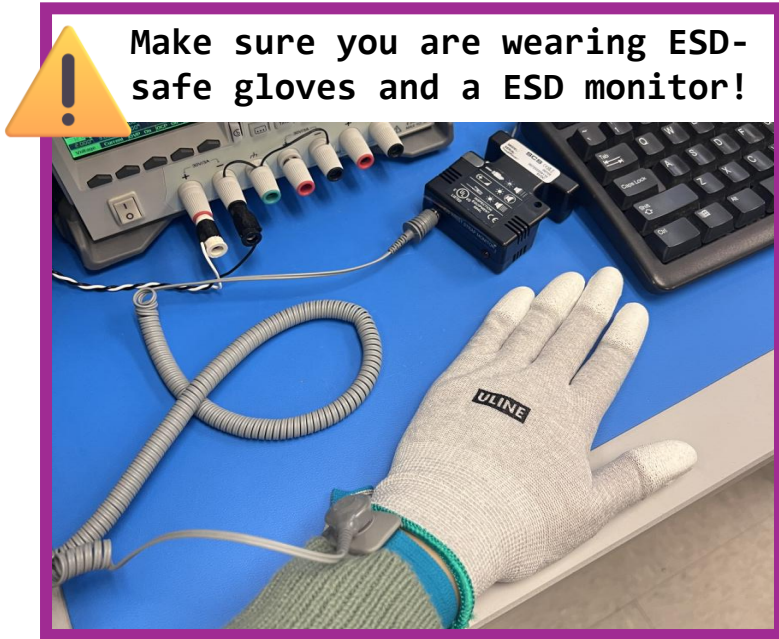


5. Insert into Bottom Slot

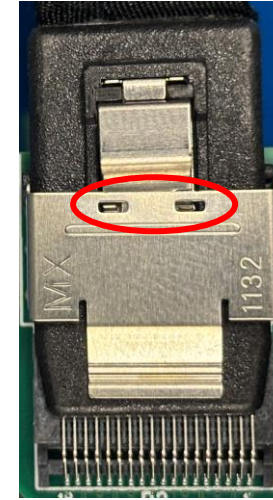
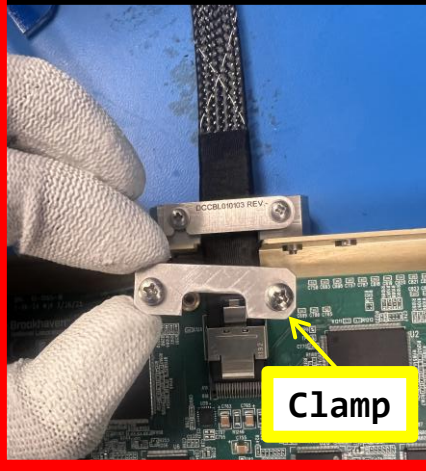


Top CE Box (VD) Assembly

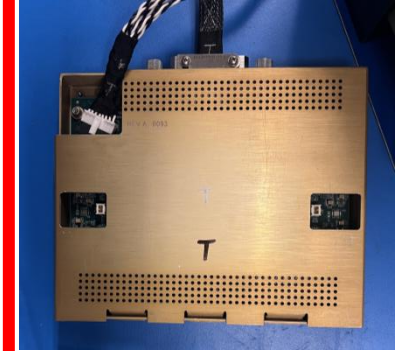
Follow these steps to assemble the Top CE Box in the support structure:



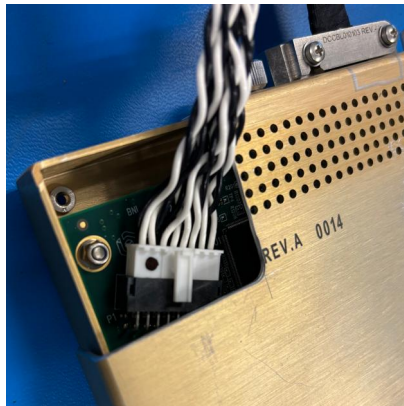
1. Install miniSAS Cable + Clamp



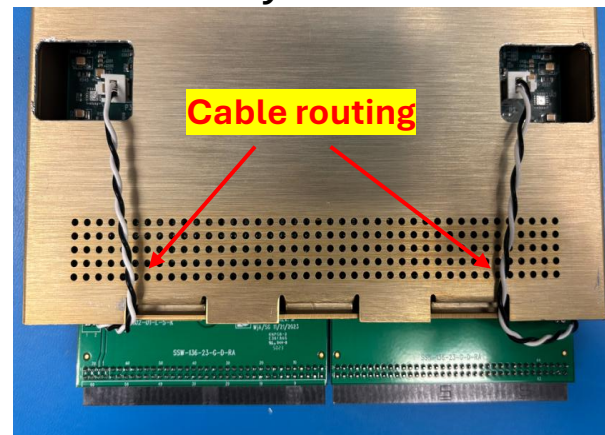
2. Install Test Cover



3. Install power cable



4. Install Toy TPCs

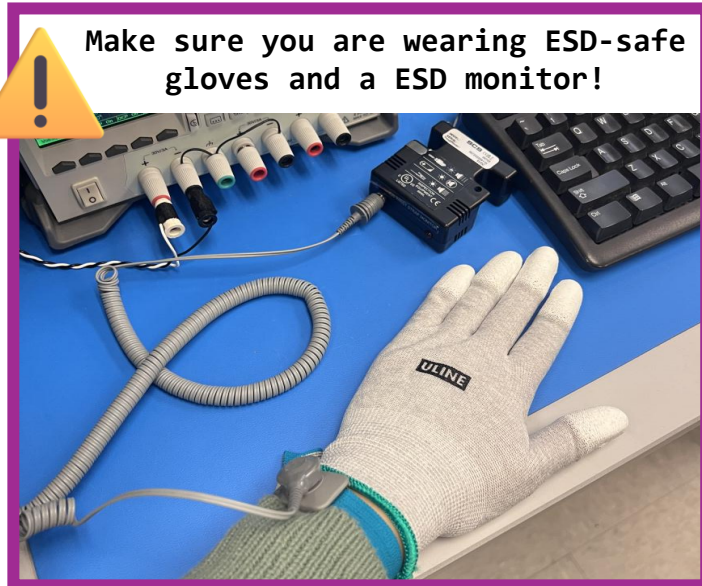


5. Insert into top Slot

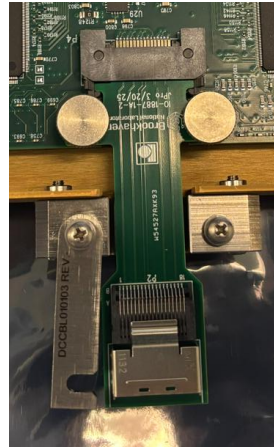


Bottom CE Box (HD) Assembly

Follow these steps to assemble the Bottom CE Box in the support structure:

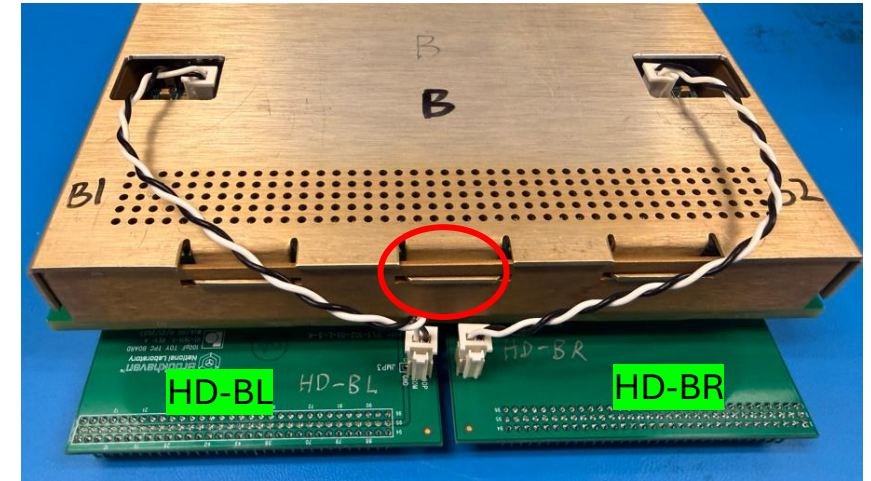


1. Install Adapter and test cover



Change later

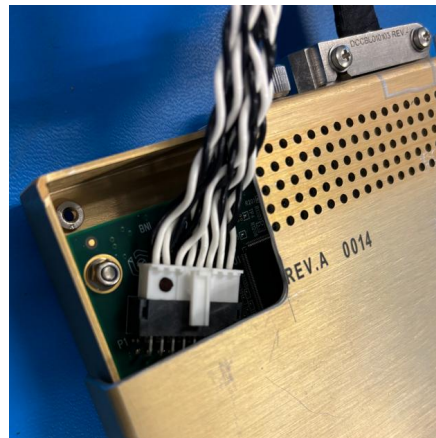
2. Remove FE terminators, install Toy TPCs



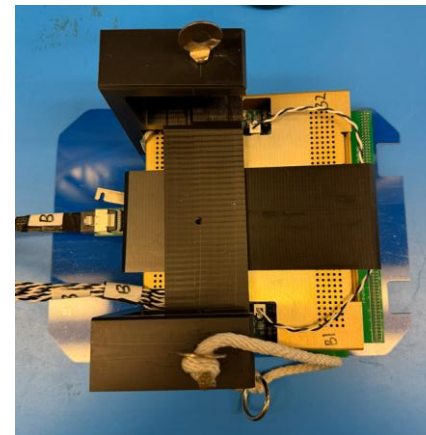
3. Install cables



Locks in position

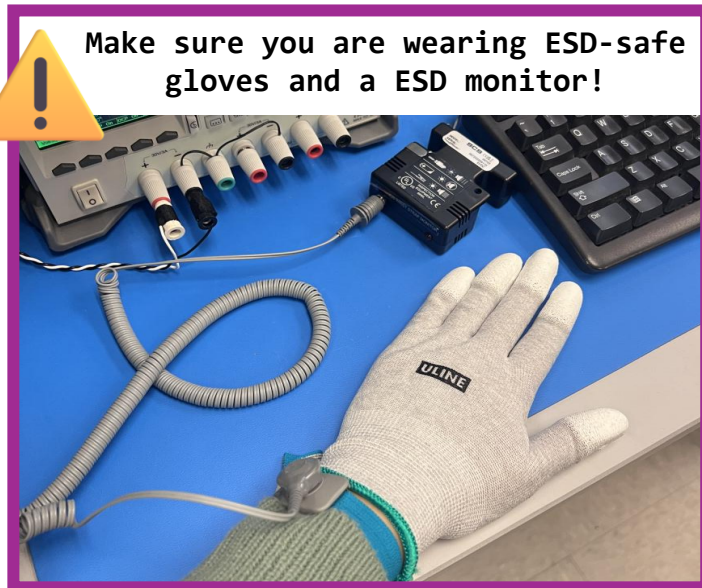


4. Insert into Bottom Slot and fasten CE box

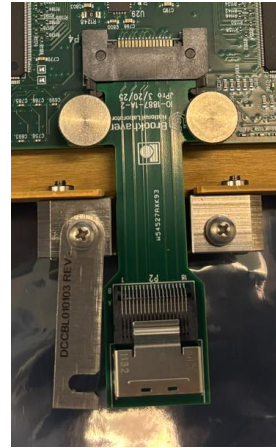


Top CE Box (HD) Assembly

Follow these steps to assemble the Bottom CE Box in the support structure:

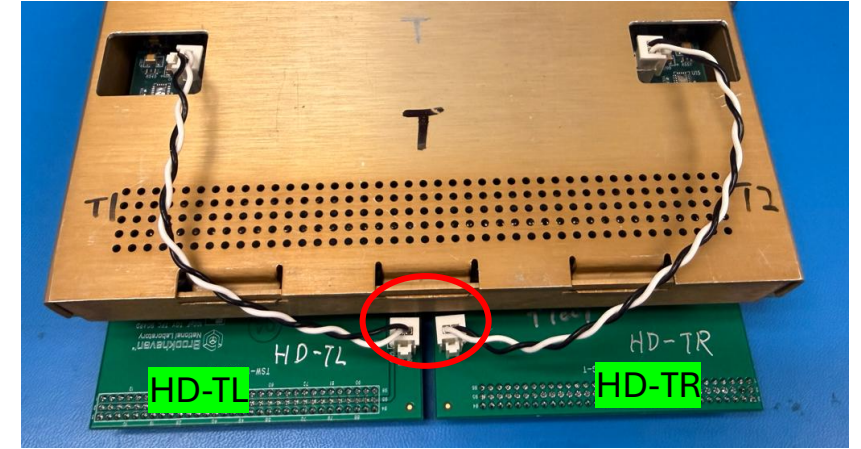


1. Install Adapter and test cover

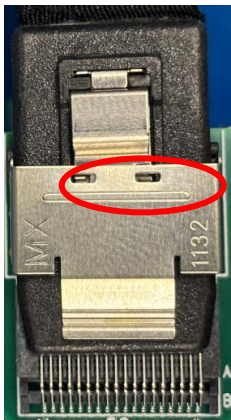


Change later

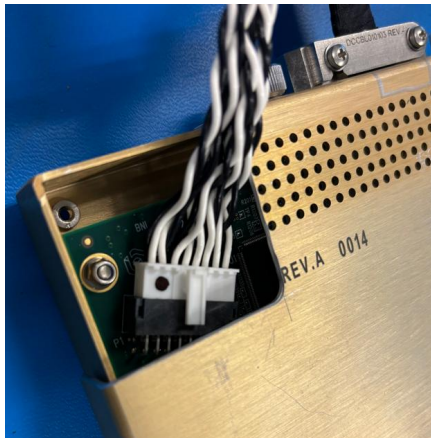
2. Remove FE terminators, install Toy TPCs



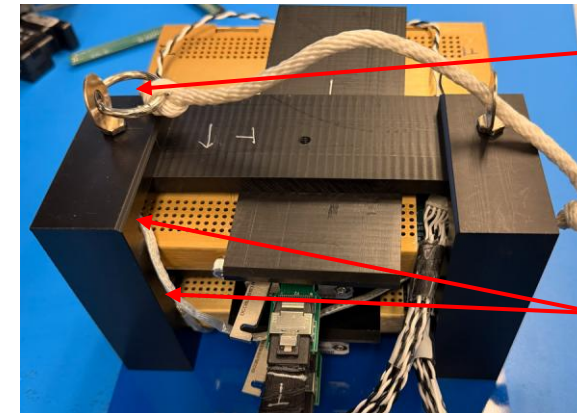
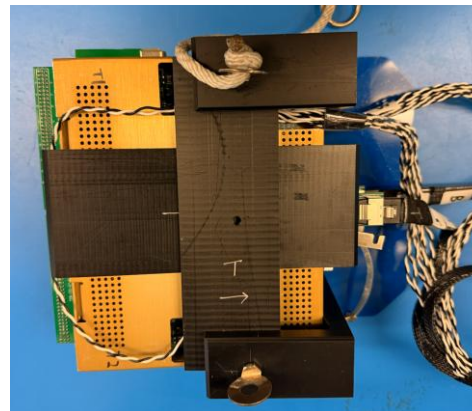
3. Install cables



Locks in position



4. Insert into top Slot and fasten CE box

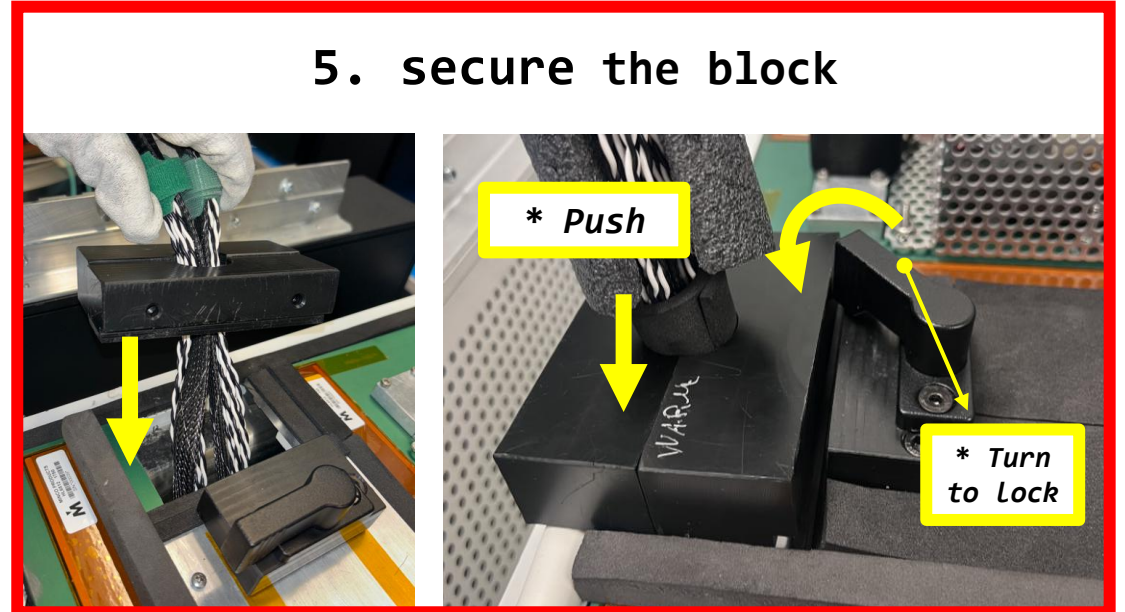
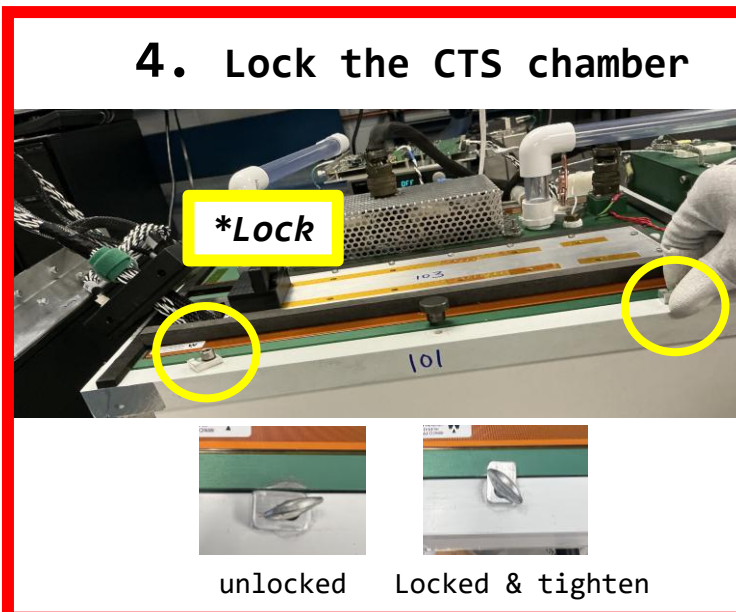
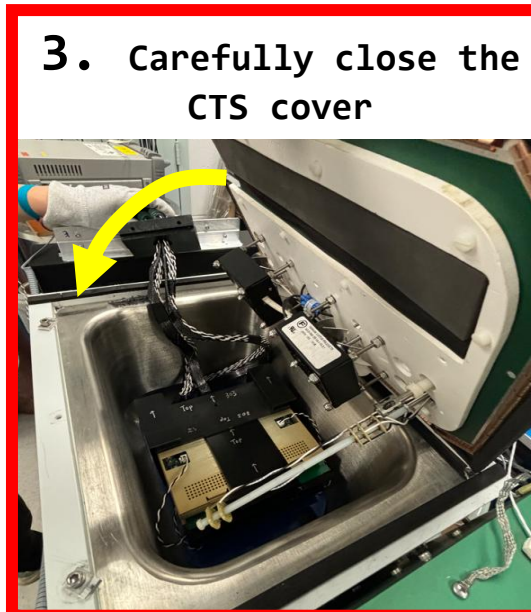
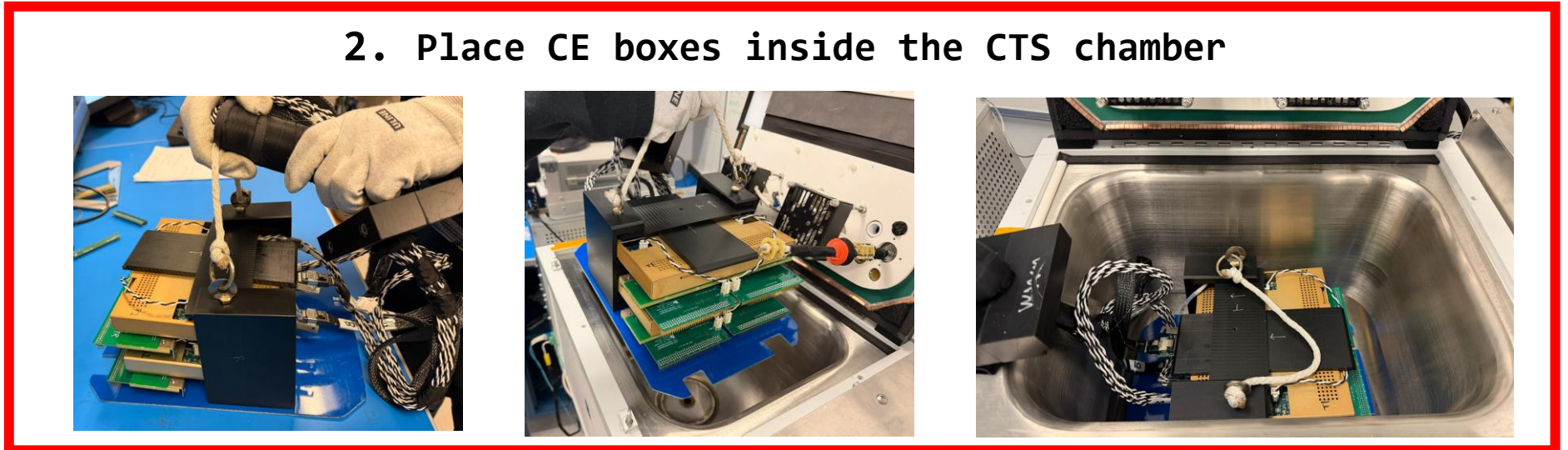
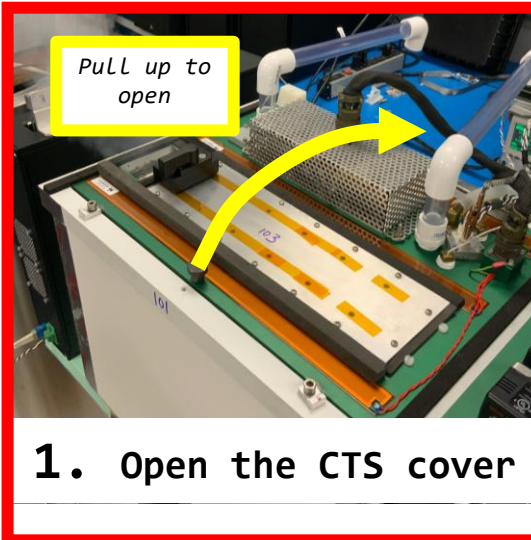


handle

Ground strips

Place CE Boxes into CTS Chamber

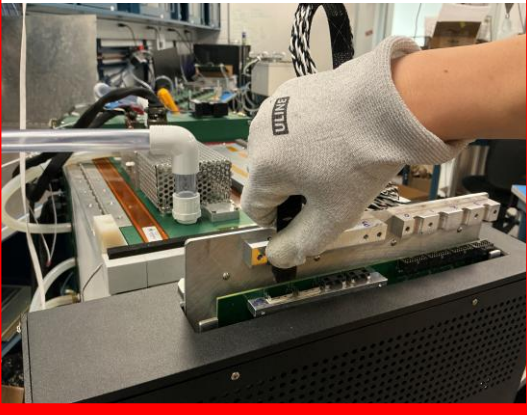
Follow these steps to install the FEMBs in the CTS:



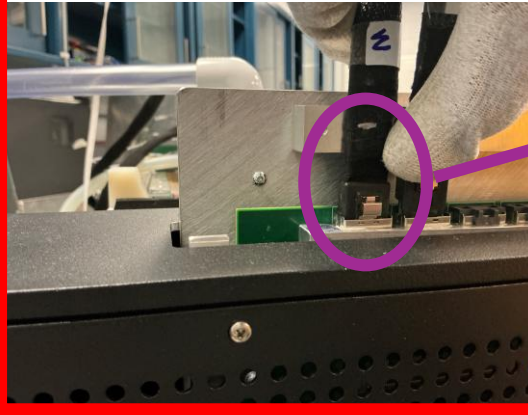
Cable Connection at WIB side

Follow these steps to connect the FEMBs to the WIB

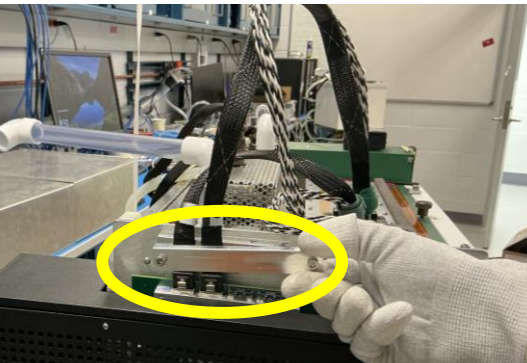
1. Place the Bottom Data Cable (B) in Slot B



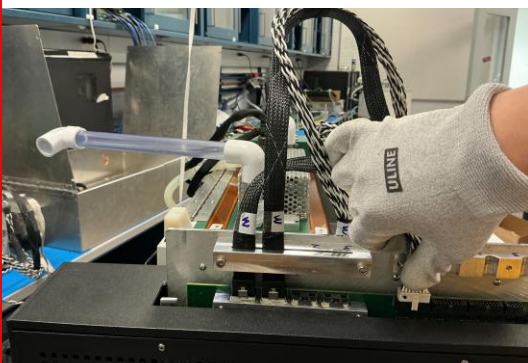
2. Place the Top Data Cable (T) in Slot T



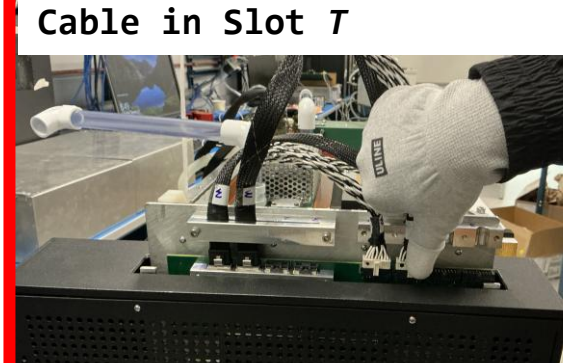
3. Install the Strain Relief Clamp



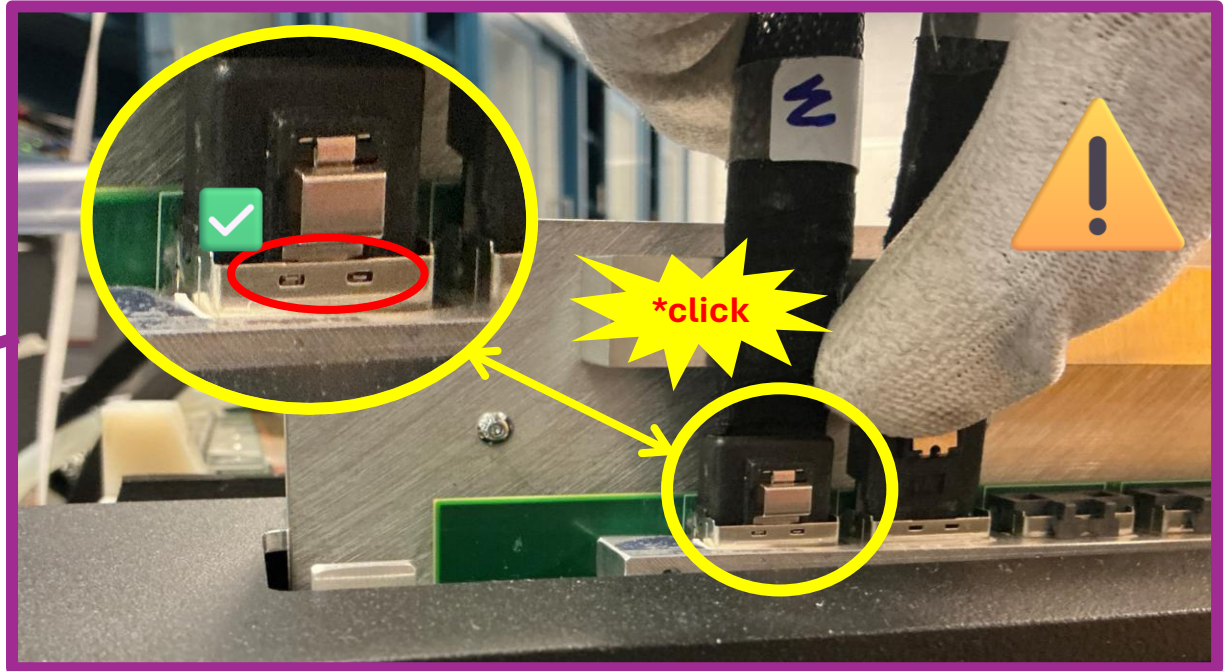
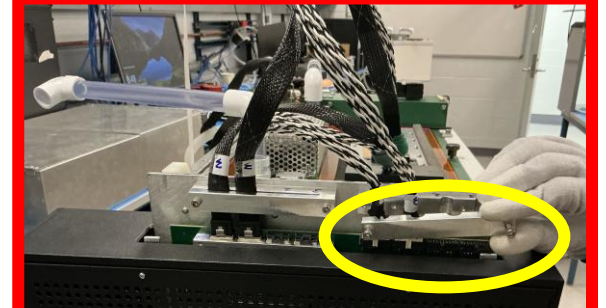
4. Install the Bottom Power Cable in Slot B



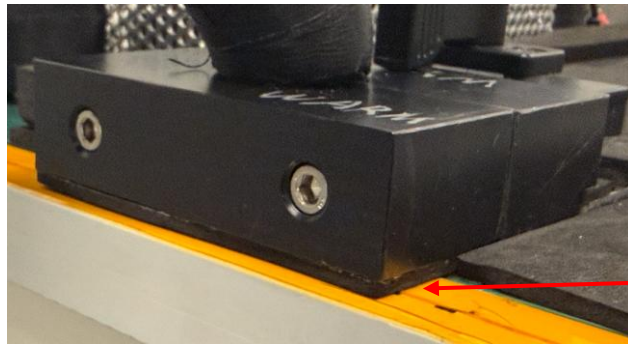
5. Install the Top Power Cable in Slot T



6. Install the Strain Relief Clamp (optional)

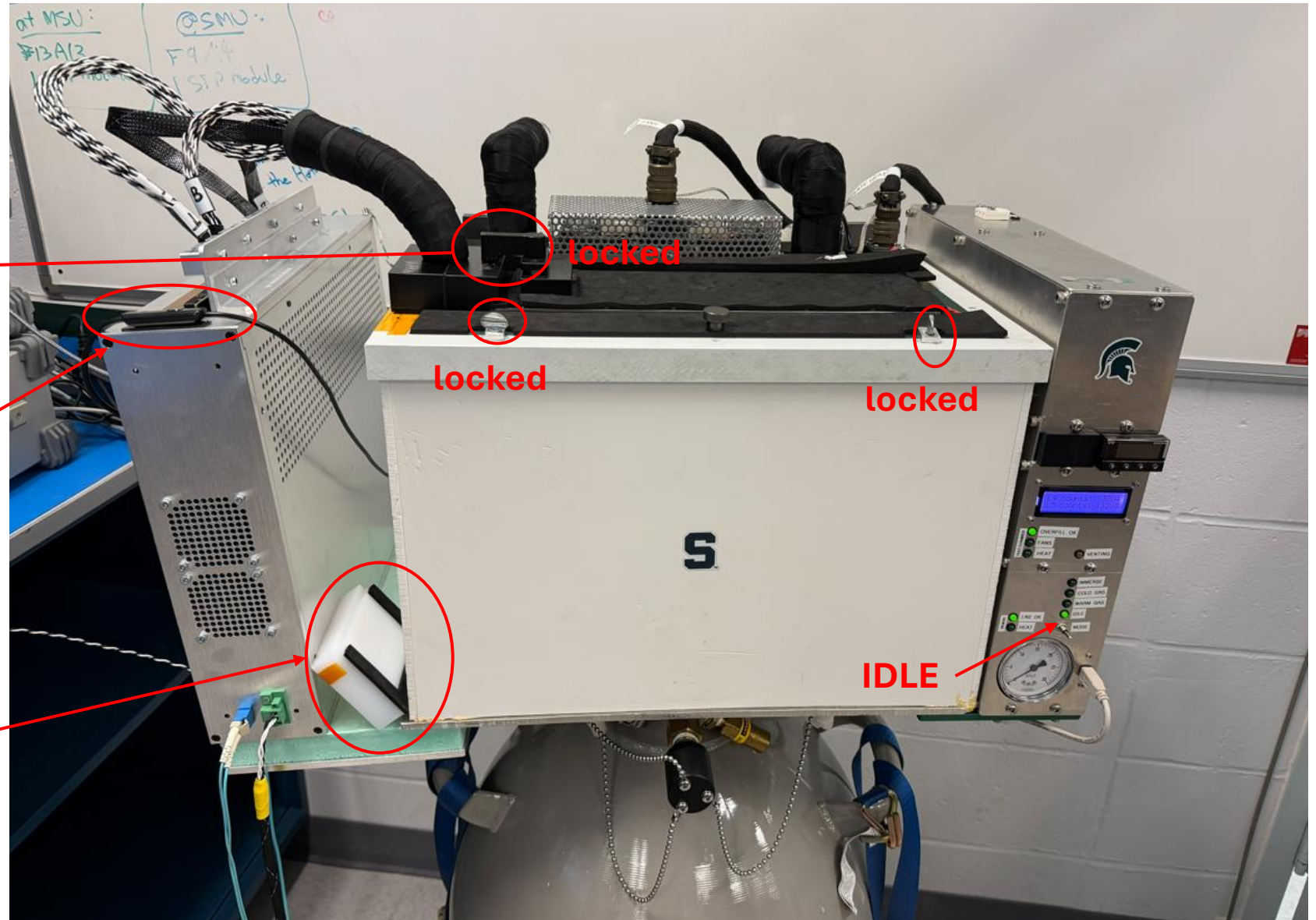


Make sure CTS is closed, locked and ready to use



No visible air gap!

Camera LED switcher



locked

locked

IDLE

When the cold cable bundle is removed, this block should seal the cover all the time!

Take CE boxes out of the Chamber

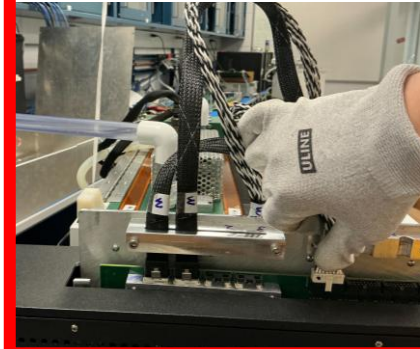
Remember to wear ESD-safe gloves and a ESD monitor!



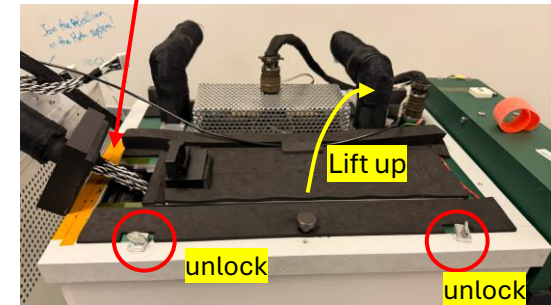
1. CTS in IDLE mode



2. Disconnect the Data and Power cables from the WIB

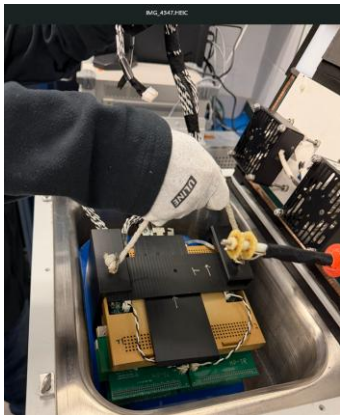


Unlock the cable bundle

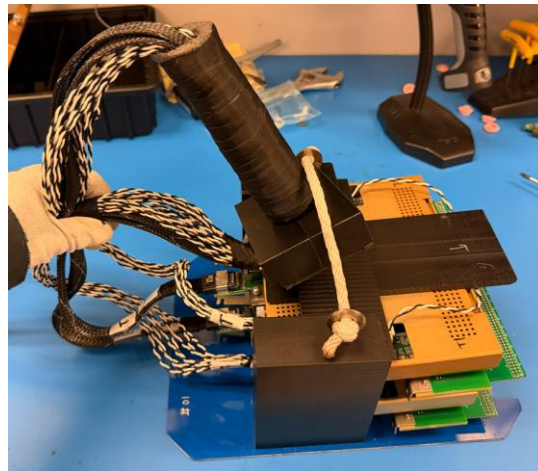


3. Open the CTS cover

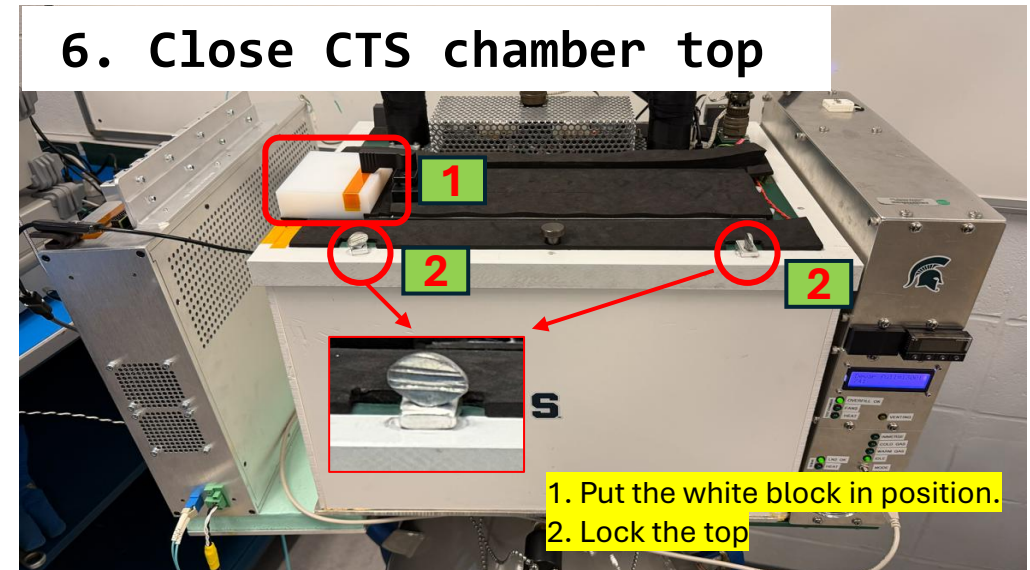
4. Hold the cable bundle, carefully lift up the CE boxes



5. Place it on the assembly area

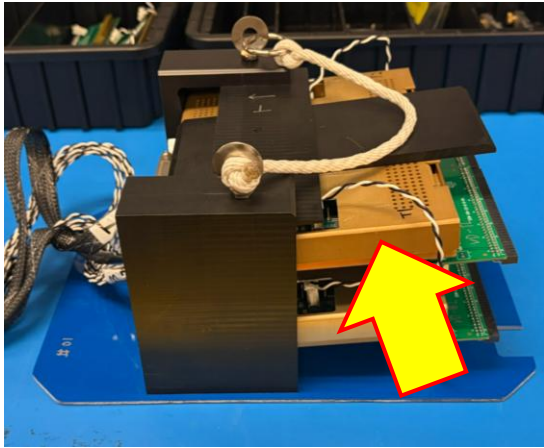


6. Close CTS chamber top

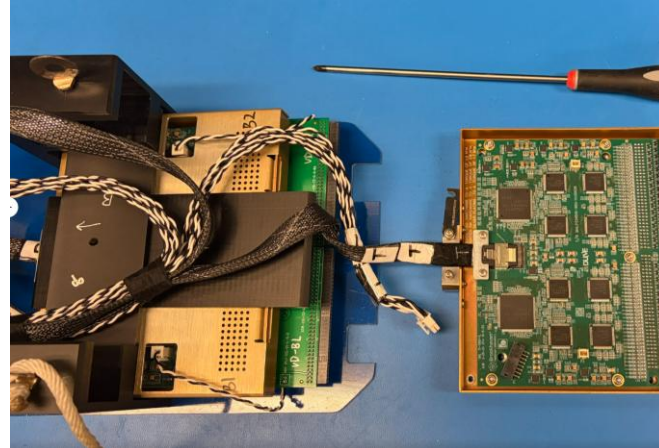


Disassembly 01: Bottom CE box (VD)

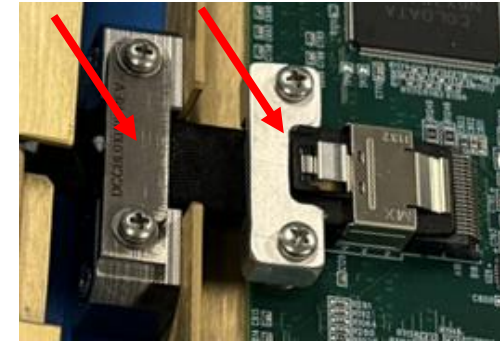
1. Remove the CE Box from the support structure



2. Remove Power Cable
3. Remove the test cover



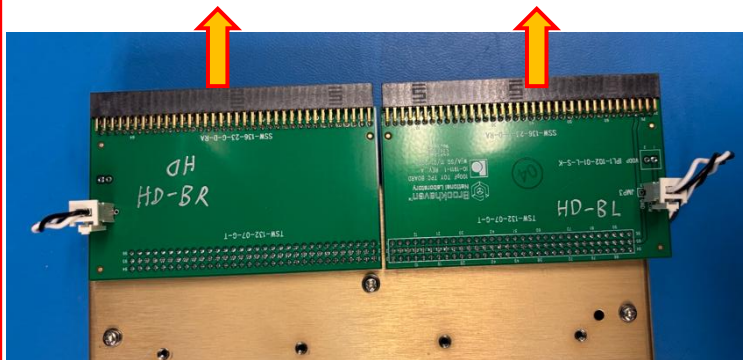
4. Remove the miniSAS cable
5. Secure connector clamp and cable relief clamp



6. Reconnect the connector clamp and cable relief clamp



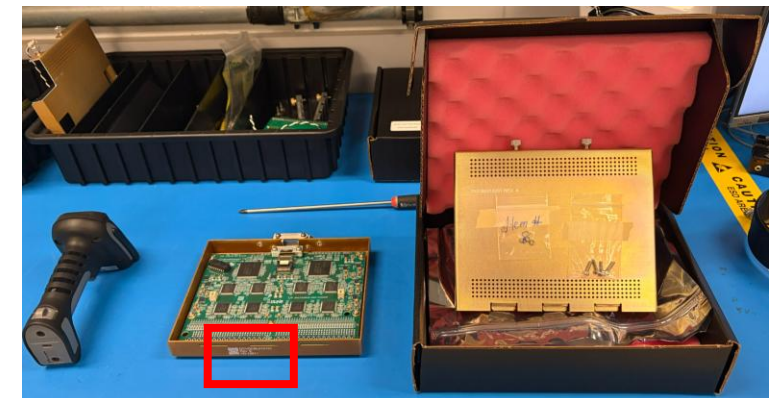
7. Remove Toy TPCs



8. Install the original CE box cover.



9. Scan CE Box ID
10. Input Cover ID

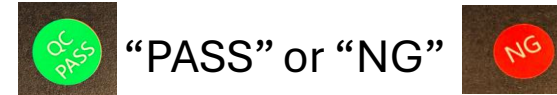


Disassembly 02: Bottom CE box (VD)

11. Pack CE box in ESD Bag and Foam Box

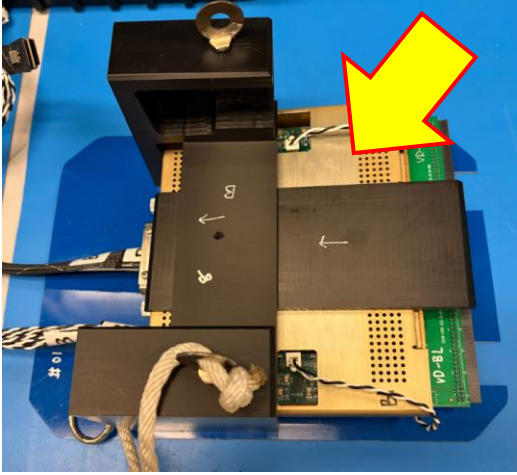


12. Scan HWDB QR code!

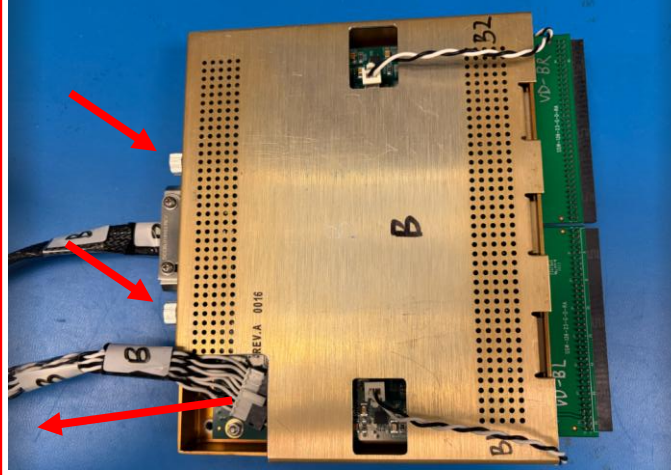


Disassembly 01: Top CE box (VD)

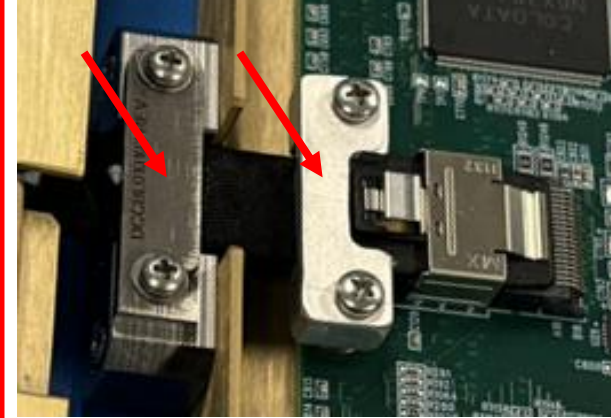
1. Remove the CE Box from the support structure



2. Remove Power Cable
3. Remove the test cover



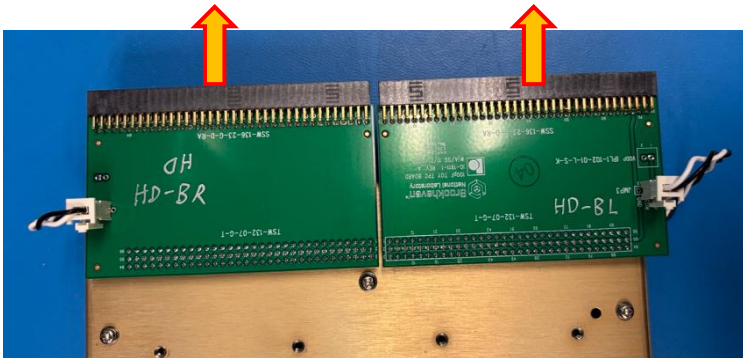
4. Remove the miniSAS cable
5. Secure connector clamp and cable relief clamp



6. Reconnect the connector clamp and cable relief clamp



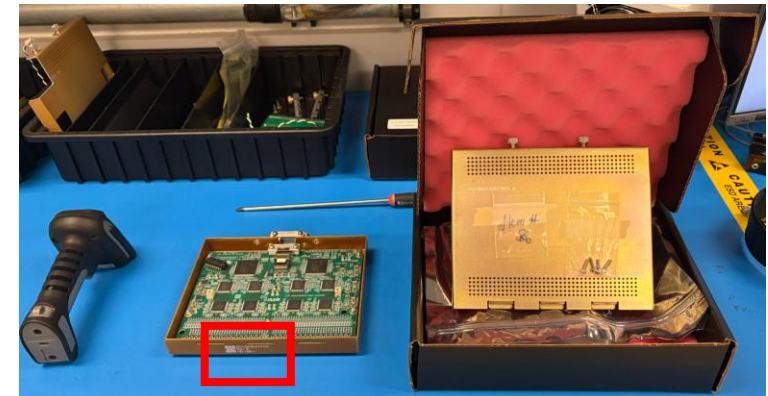
7. Remove Toy TPCs



8. Install the original CE box cover.



9. Scan CE Box ID
10. Input Cover ID

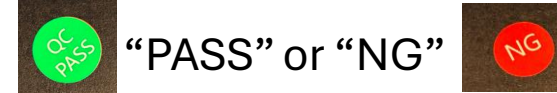


Disassembly 02: Top CE box (VD)

11. Pack CE box in ESD Bag and Foam Box



12. Scan HWDB QR code!

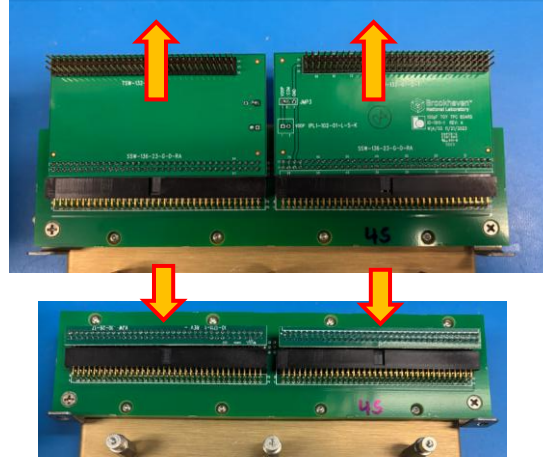


Disassembly 01: CE box (HD)

1. Remove the CE Box from the support structure
2. Remove Power Cable
3. Remove the test cover

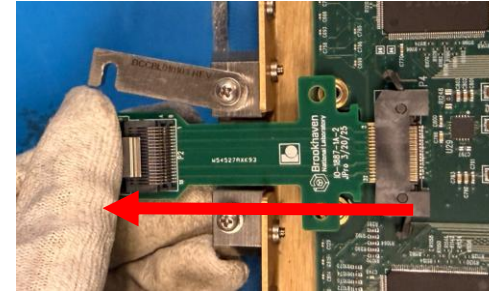


4. Remove Toy TPCs and install FE terminators back



5. unscrew, Slide out the miniSAS adapter board

6. Secure cable relief clamp



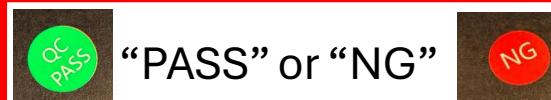
7. Install the original CE box cover.
Scan CE box QR code.
Entry CE box cover SN last 4 digits.



8. Pack CE box in ESD Bag and Foam Box



Scan HWDB QR code!



Store it!

Put Accessories back to original positions

Put accessories back to tray!



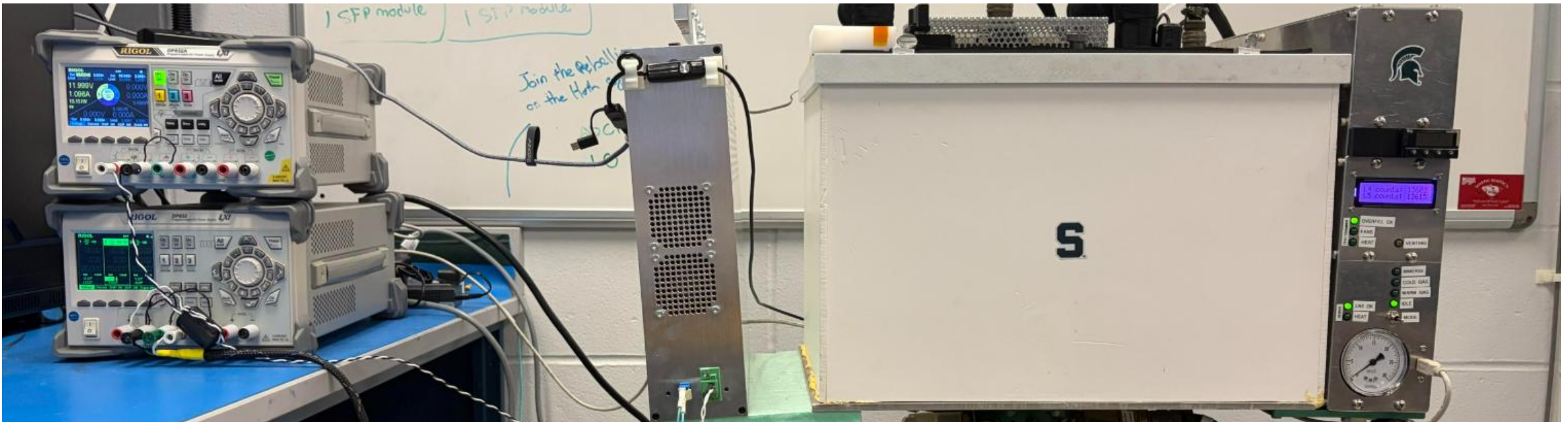
Maintain the assembly area neat and clean!



1: Assembly area

2: Place the assembled CE box support structure (to be tested in the next run)

Is the assembly area neat and clean? Confirm the CTS back to normal

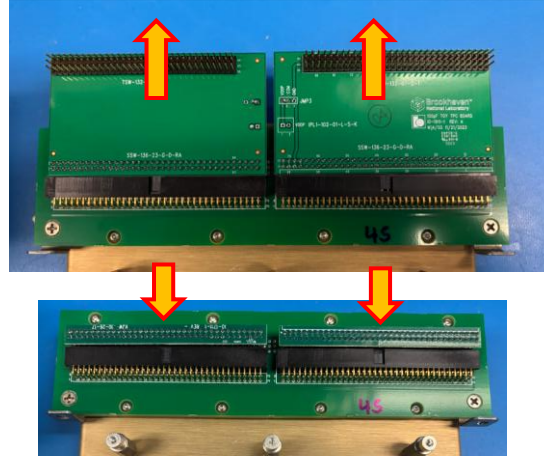


Disassembly: CE box (HD)

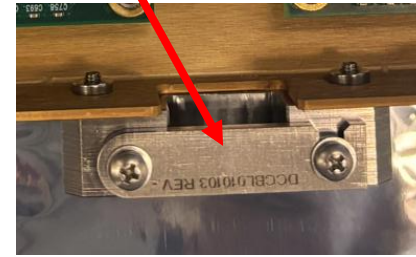
- 1.** Remove the CE Box from the support structure



- 2.** Remove Toy TPCs and install FE terminators back



- 3.** Remove Power Cable
4. Remove the test cover
5. Remove the miniSAS cable and its adapter board
6. Secure cable relief clamp



- 7.** Install the original CE box cover



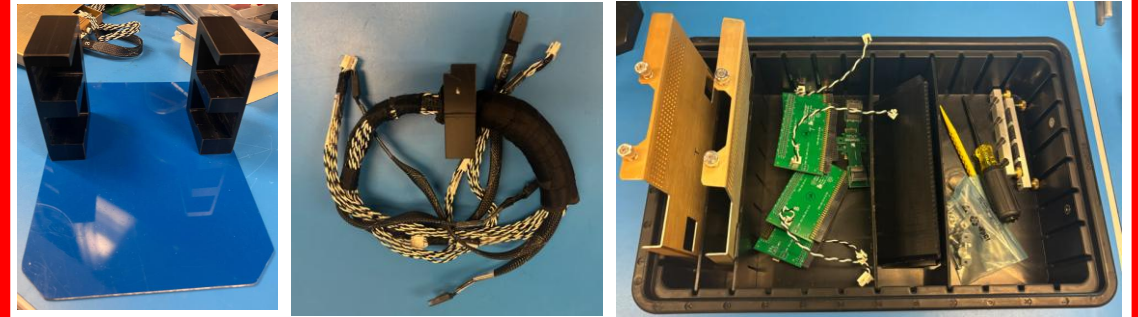
- 8.** Pack CE box in ESD Bag and Foam Box



“PASS” or “NG”



- 9.** Put accessories back to tray!



How to convert each slide in a PowerPoint (.pptx) file into a .png image file — one image per slide.

<https://drive.google.com/file/d/1LLRxvgbRMVB5utvIPry4vlhtx9JX7j-2/view?usp=sharing>

If you're on **Windows** and have **Microsoft PowerPoint** installed, you can use the `comtypes` automation library to export slides directly to images.

backup

Initial Checkout List (FEMB-VD)

Locate the following items before starting:



Safety First!



Long
Sleeve Top

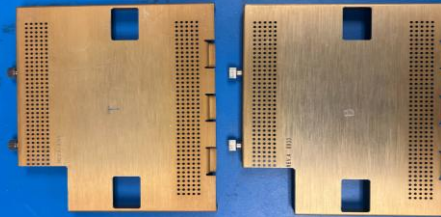
Long Pants

Closed Toe
Shoes

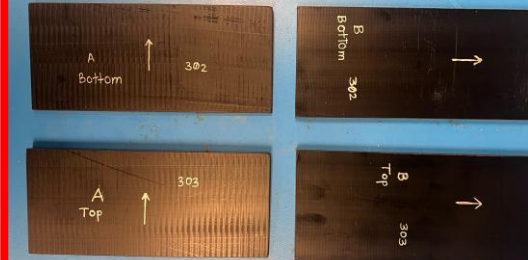
ODH Monitor



Top & Bottom Test
Covers



Top & Bottom Clamps



Data cable
clamps (x2)



Cryogenic Gloves



ESD-safe Gloves



Anti-static
Wrist Strap

Cryogenic
Face Mask

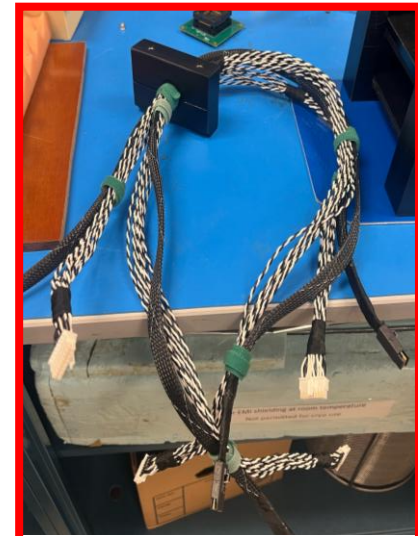


Support Structure

Toy TPC with Cables
(2 pairs)



Lingyun: Replace with new picture



1m Cable Bundle

Lingyun: Replace with new picture

Initial Checkout List (FEMB-HD)

Locate the following items before starting:



Safety First!



Long
Sleeve Top

Long Pants

Closed Toe
Shoes

ODH Monitor



ESD-safe Gloves

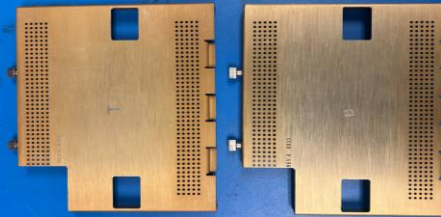
Cryogenic Gloves



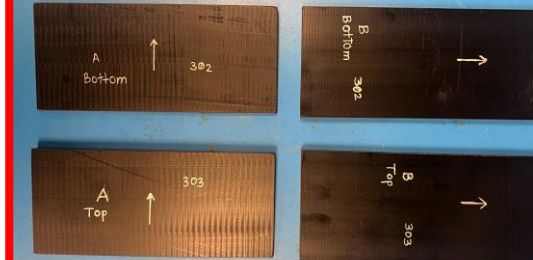
Anti-static
Wrist Strap

Cryogenic
Face Mask

Top & Bottom Test
Covers



Top & Bottom Clamps



Data cable
clamps (x2)



Lingyun: MiniSAS cable
adapter and thumb-nuts

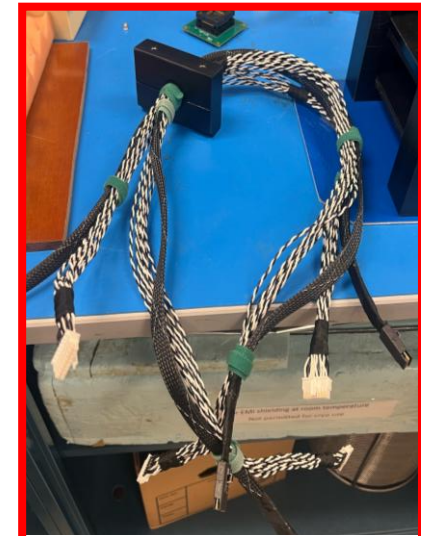


Support Structure

Toy TPC with Cables
(2 pairs)



Lingyun: Replace with new picture

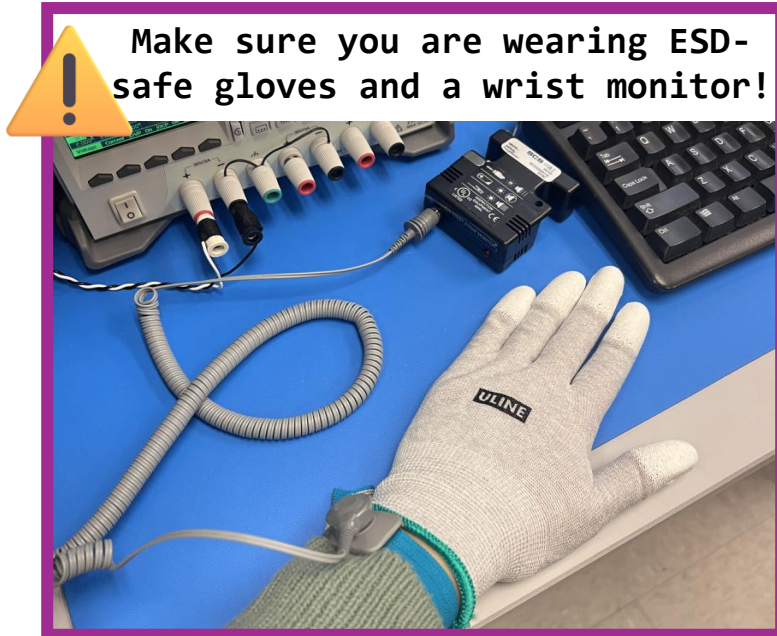


1m Cable Bundle

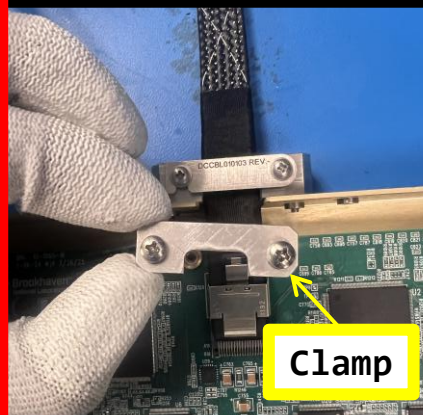
Lingyun: Replace with new picture

Bottom CE Box (VD) Assembly

Follow these steps to assemble the Bottom CE Box in the support structure:



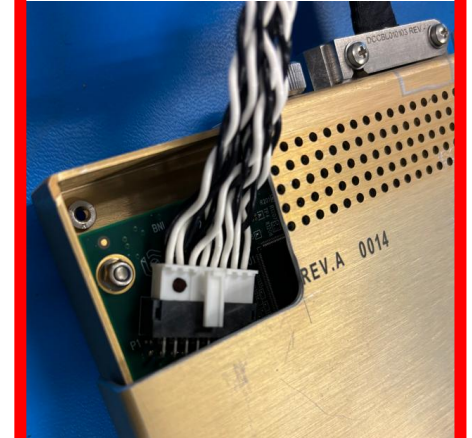
1. Install the Signal Cable + Clamp



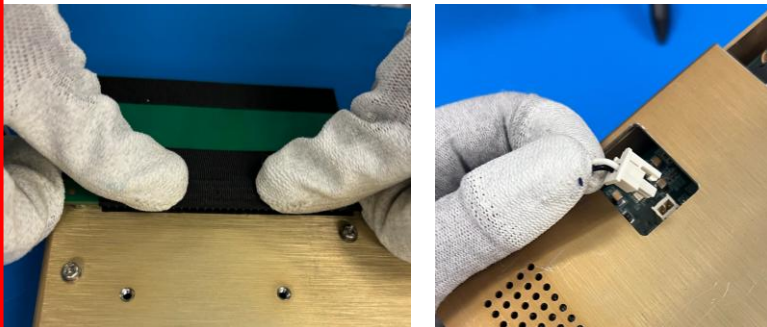
2. Install Test Cover



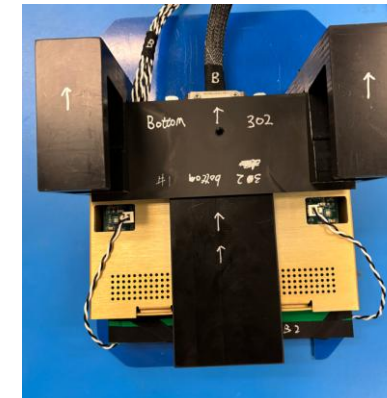
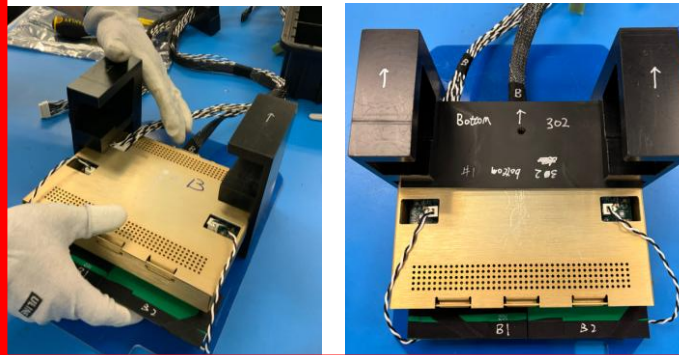
3. Install Power Cable



4. Connect Toy TPCs



5. Insert into Bottom Slot

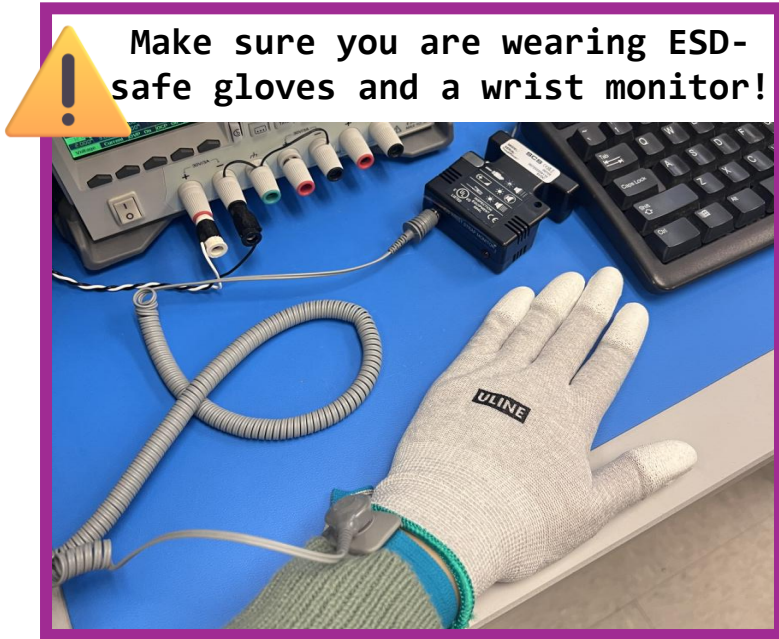


Done

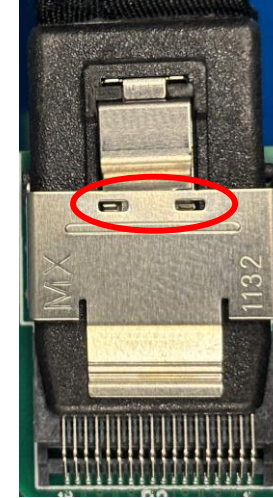
@Lingyun, Provide the routing of cables between ToyTPCs and FEMB?

Bottom CE Box (VD) Assembly

Follow these steps to assemble the Bottom CE Box in the support structure:



1. Install miniSAS Cable + Clamp

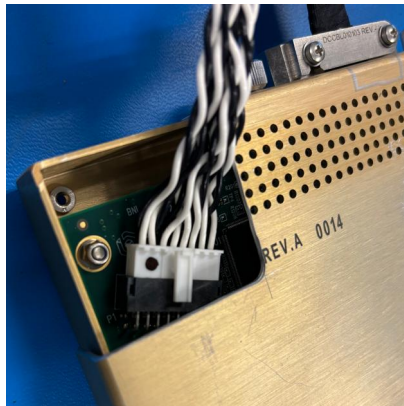


Locks in position

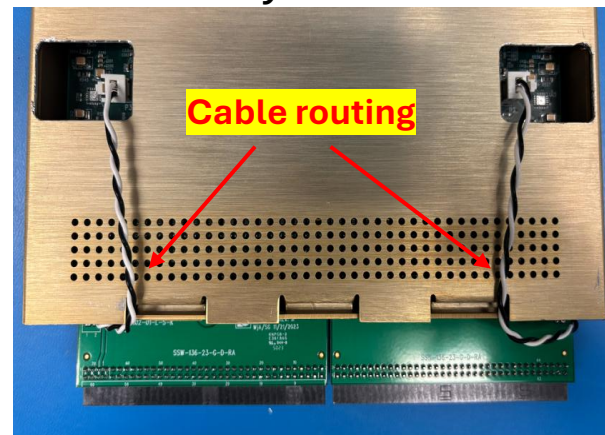
2. Install Test Cover



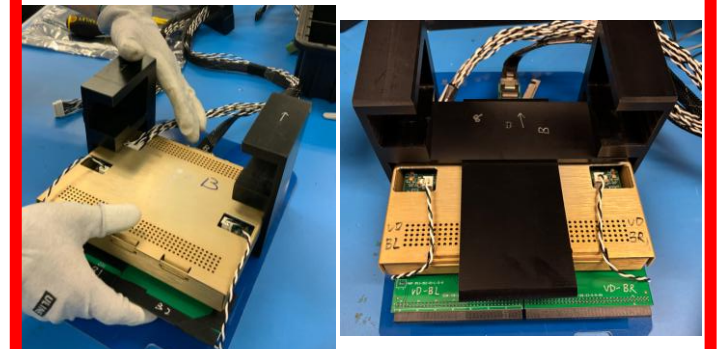
2. Install power cable



4. Install Toy TPCs



5. Insert into Bottom Slot



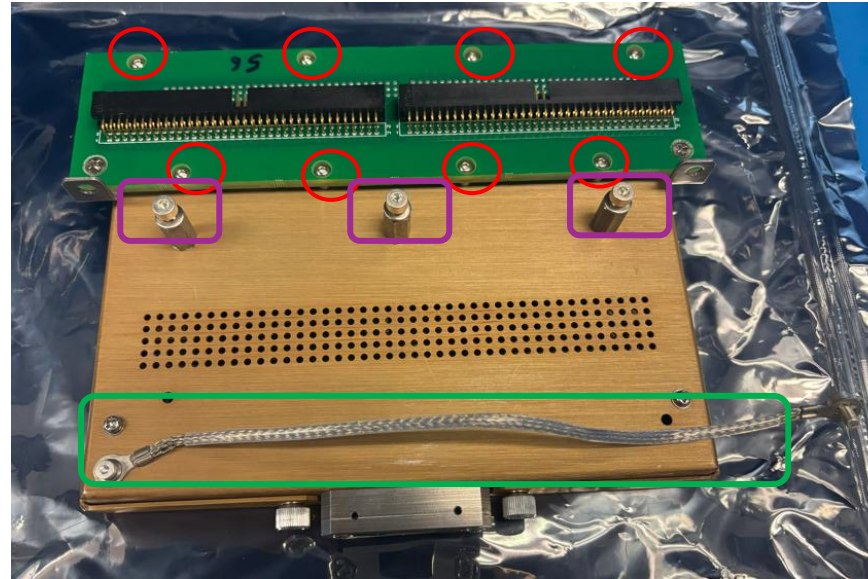
Bottom CE Box (HD) Visual Inspection



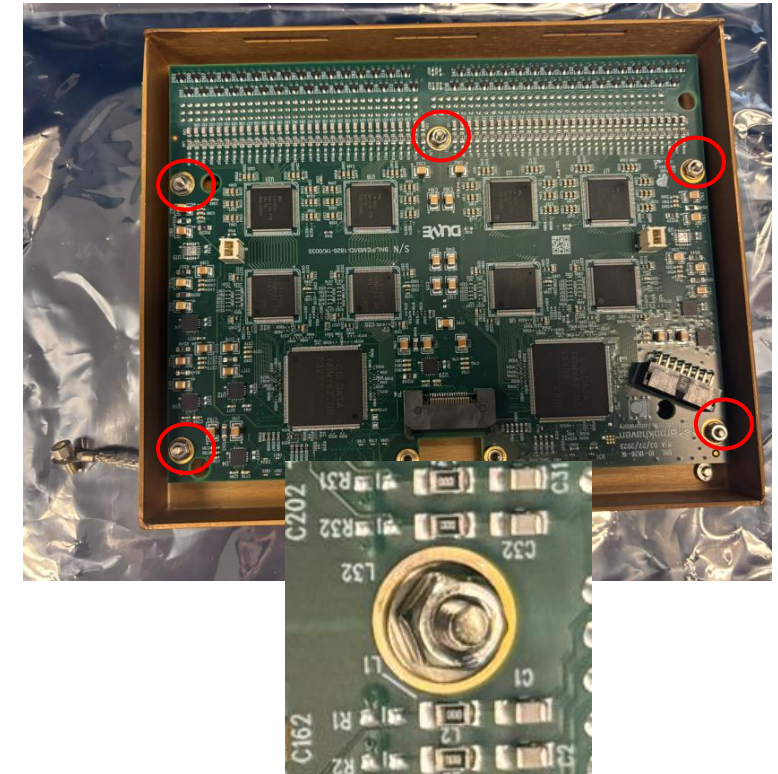
If foam box was broken?



CE box enclosed by ESD bag?



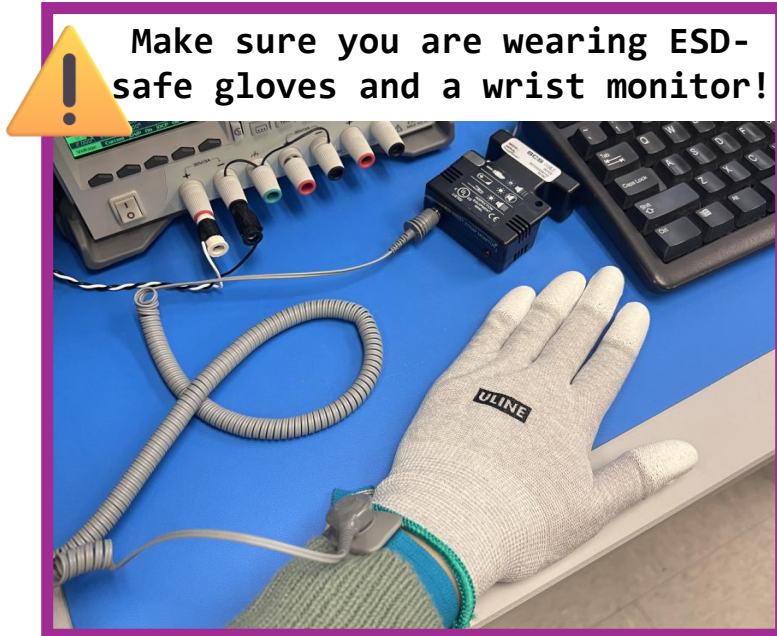
Nuts for CR-RC adapter board in position?
3 sets of lock nut and washer in position?
Ground strip in position?



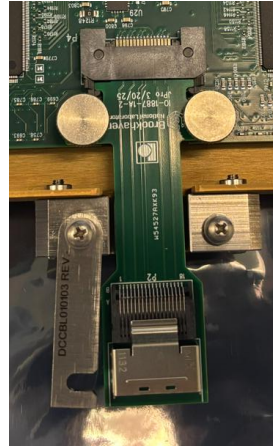
Are these 5 nuts fastened with washers?

Bottom CE Box (HD) Assembly

Follow these steps to assemble the Bottom CE Box in the support structure:

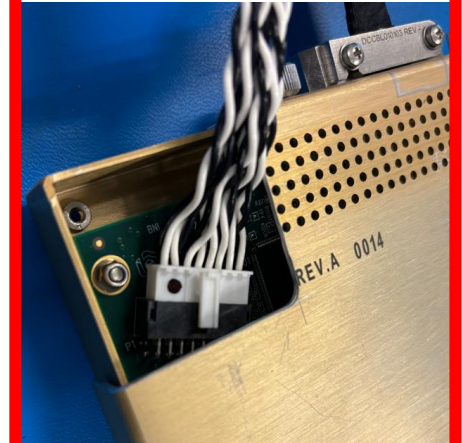


1. Install Adapter + miniSAS Cable



2. Install Test Cover
3. Install Power Cable

3. Install Power Cable



4. Connect Toy TPCs

Update for HD

5. Insert into Bottom Slot

Update for HD

Update for HD

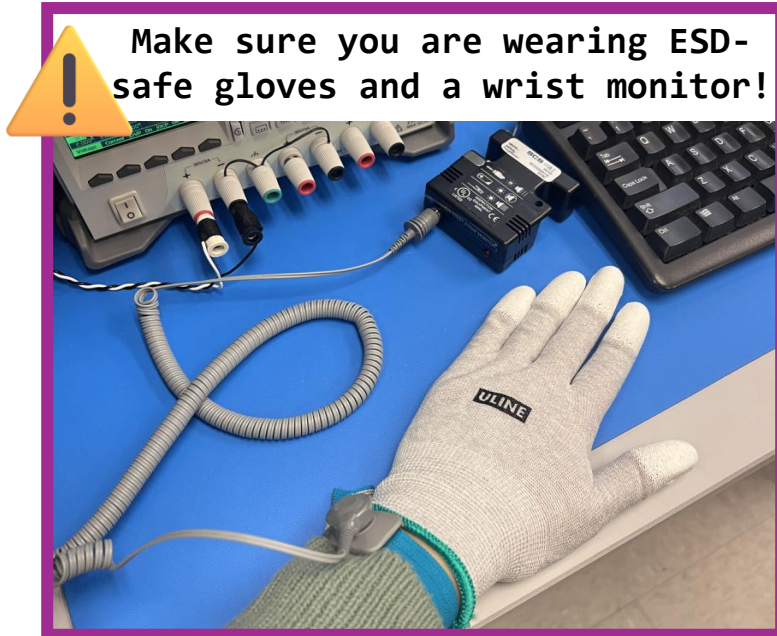


Done

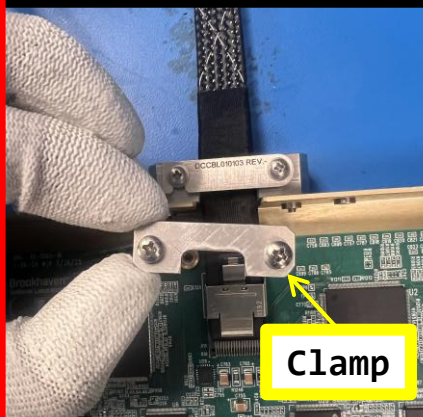
@Lingyun, Provide the routing of cables between ToyTPCs and FEMB?

Top CE Box (VD) Assembly

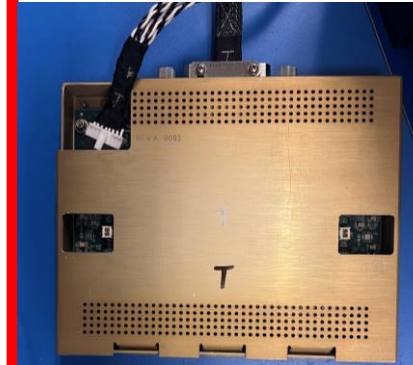
Follow these steps to assemble the Top CE Box in the support structure:



1. Install the
Signal Cable +
Clamp

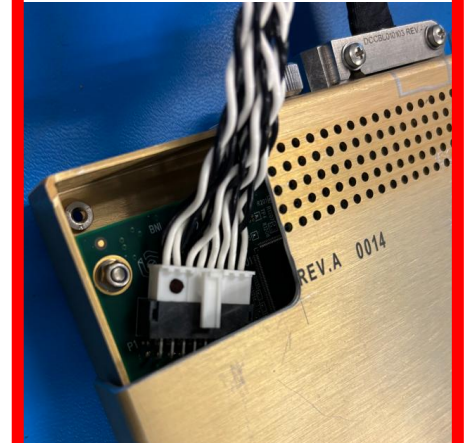


2. Install
Test Cover

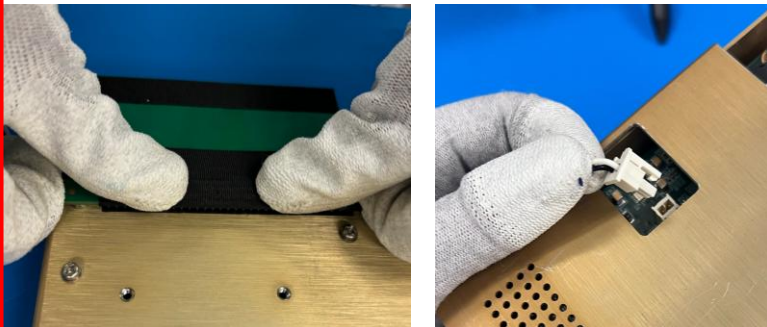


Lingyun: Replace the picture

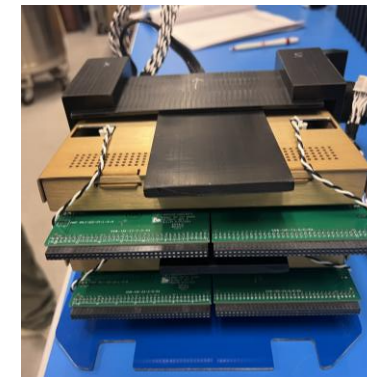
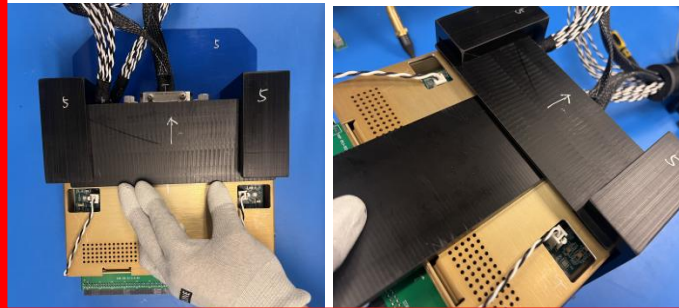
3. Install Power
Cable



4. Connect Toy TPCs



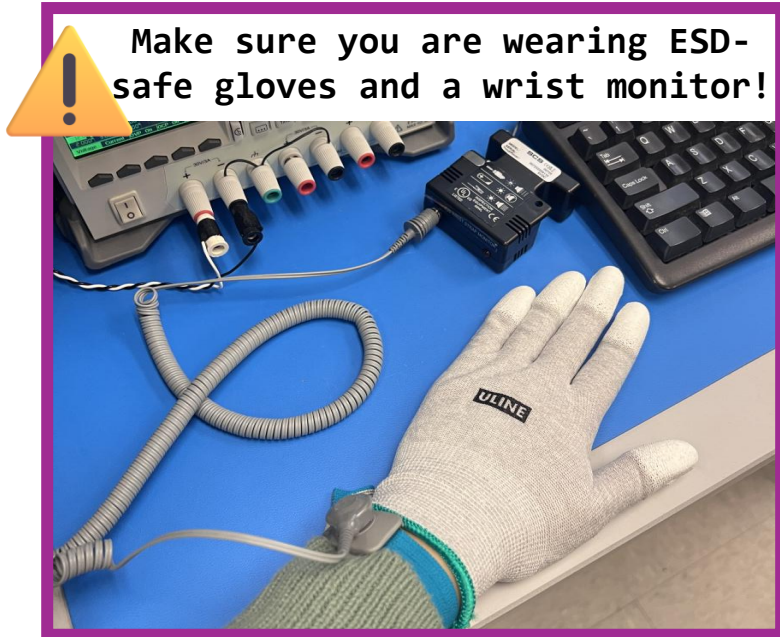
5. Insert into Bottom Slot



Done

Top CE Box (HD) Assembly

Follow these steps to assemble the Top CE Box in the support structure:

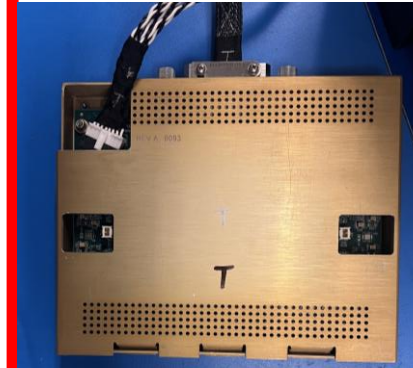


1. Install the
Signal Cable +
Clamp

Update for HD



2. Install
Test Cover



Lingyun: Replace the picture

3. Install Power
Cable



4. Connect Toy TPCs

Update for HD

5. Insert into Bottom Slot

Update for HD

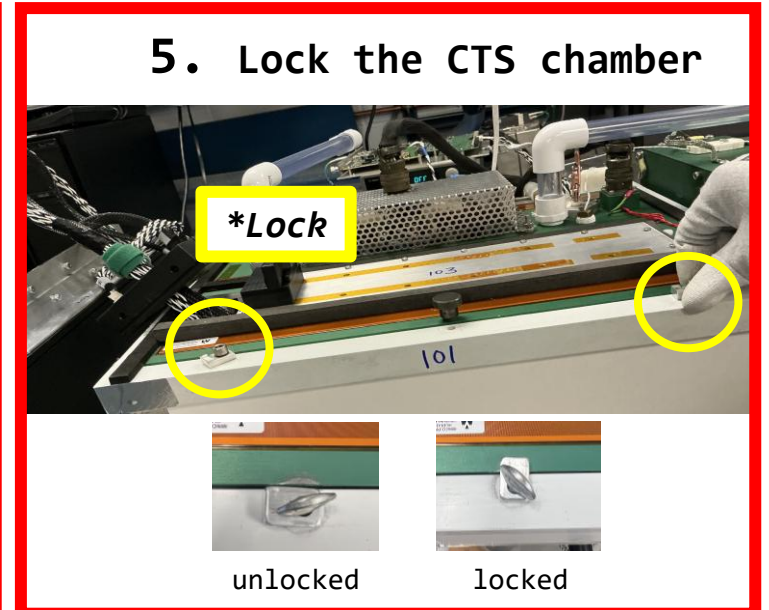
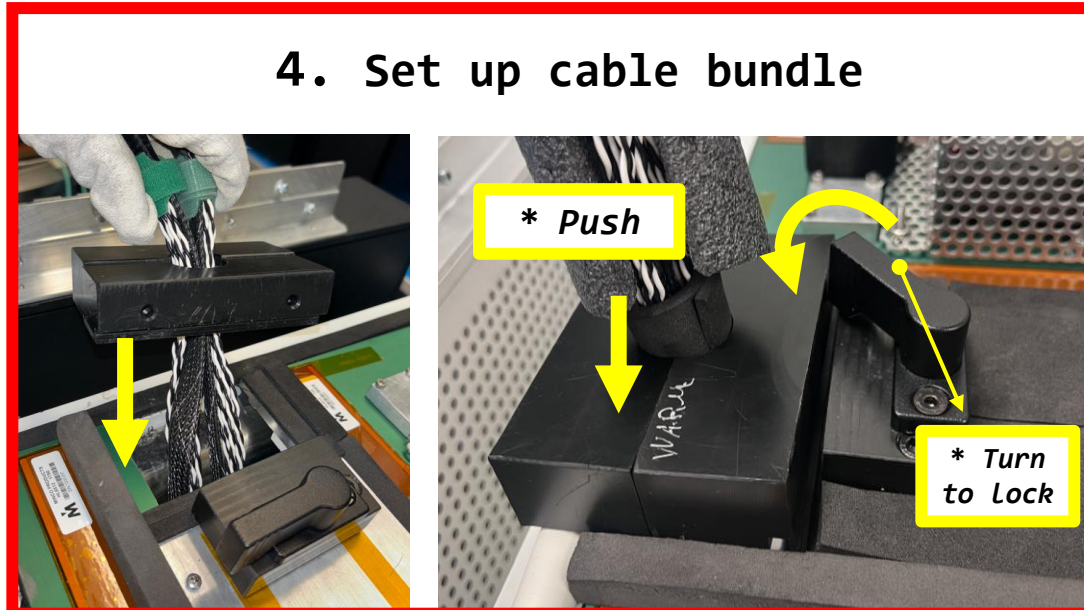
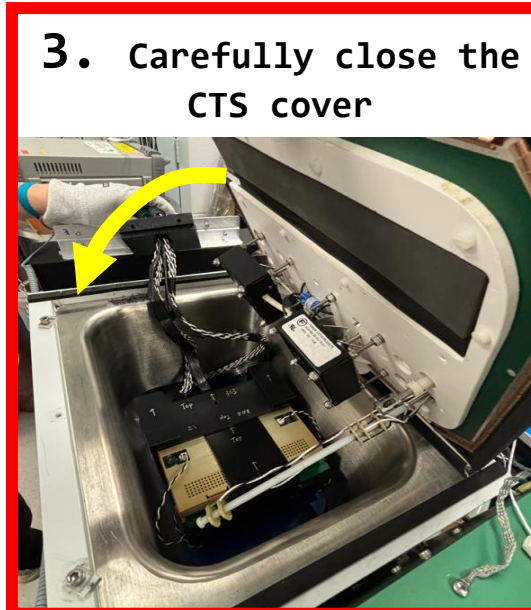
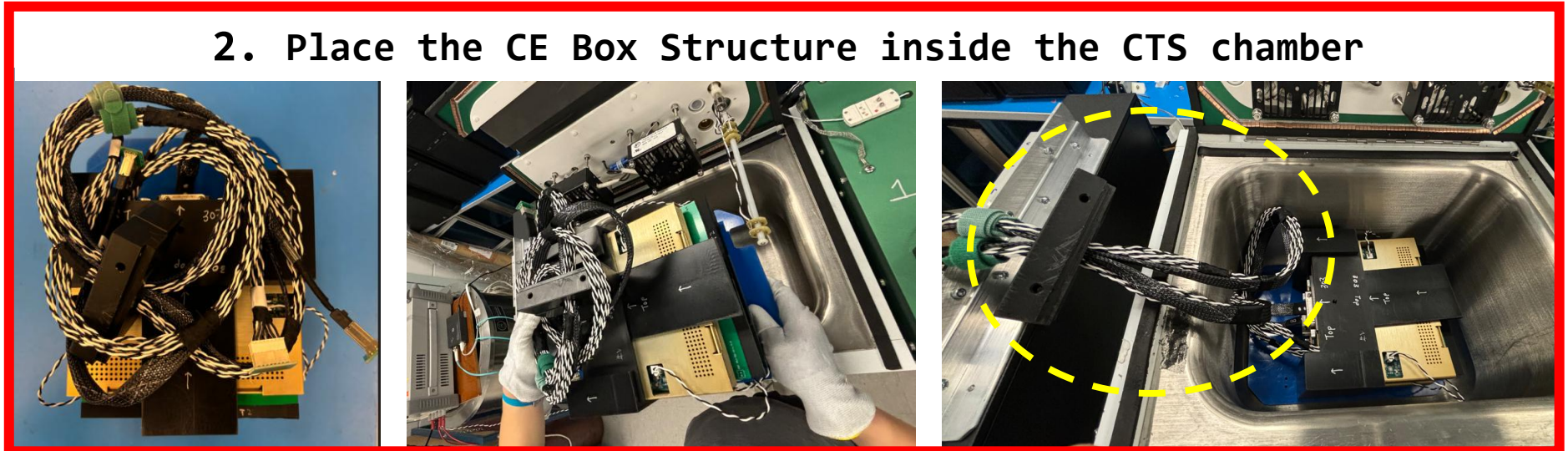
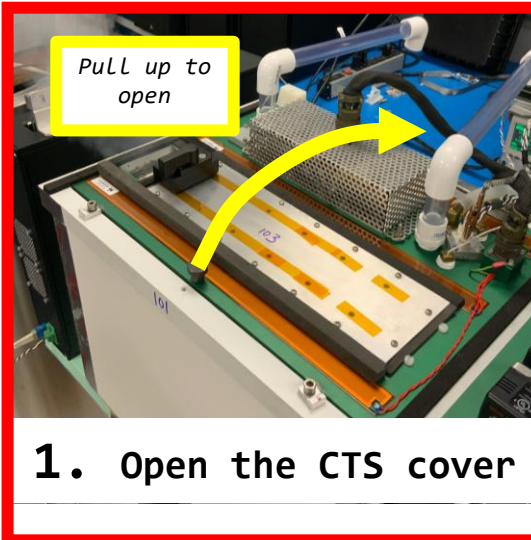
Update for HD



Done

Place CE Boxes into CTS Chamber

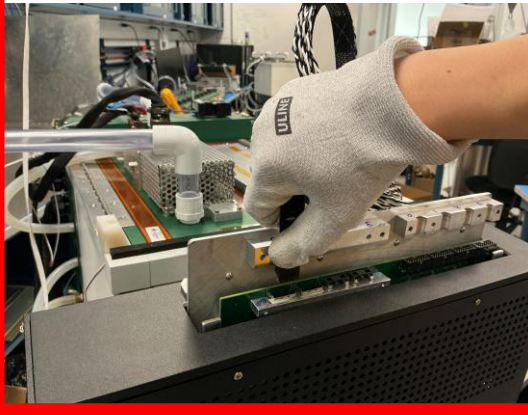
Follow these steps to install the FEMBs in the CTS:



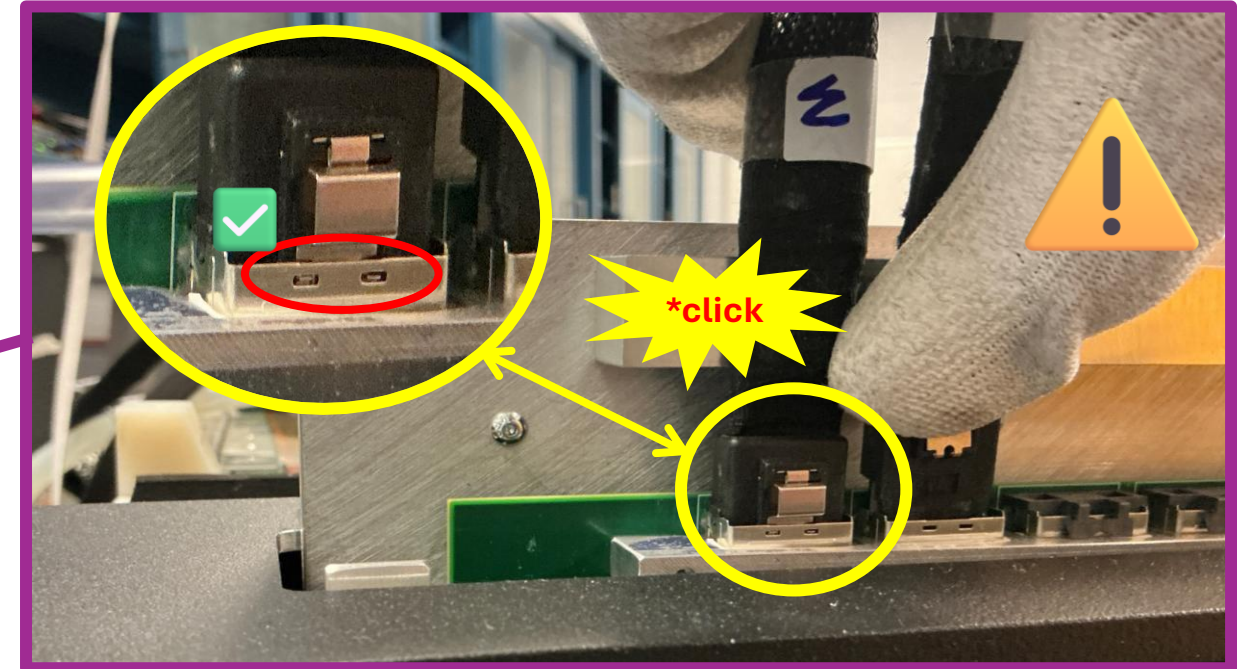
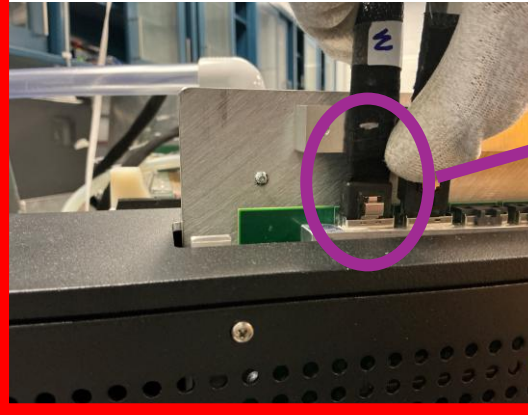
WIB Cable Connection

Follow these steps to connect the FEMBs to the WIB

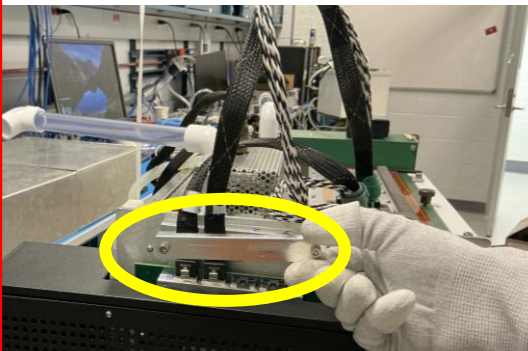
1. Place the Bottom Data Cable in Slot B



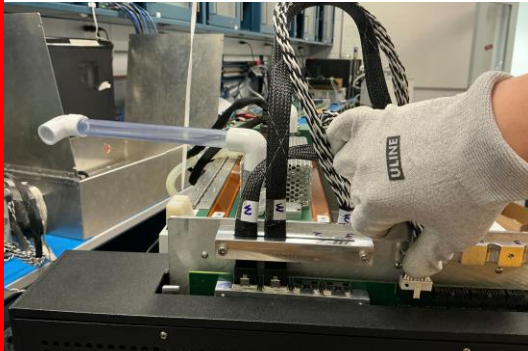
2. Place the Top Data Cable in Slot T



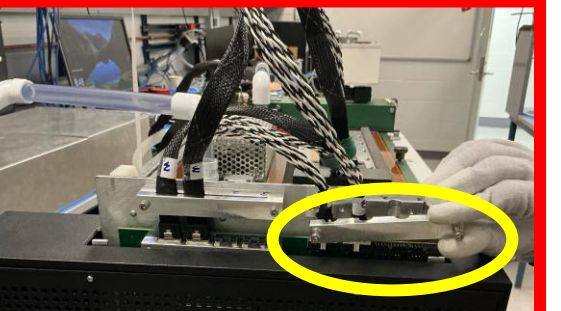
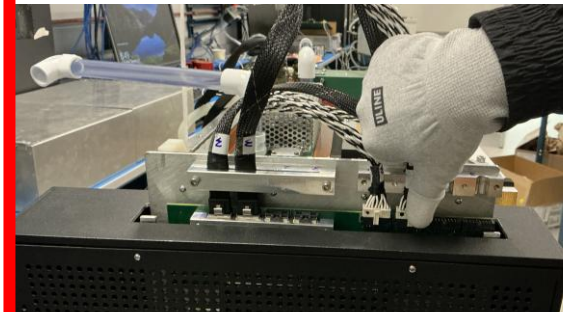
3. Install the Strain Relief Clamp



4. Install the Bottom Power Cable in Slot B



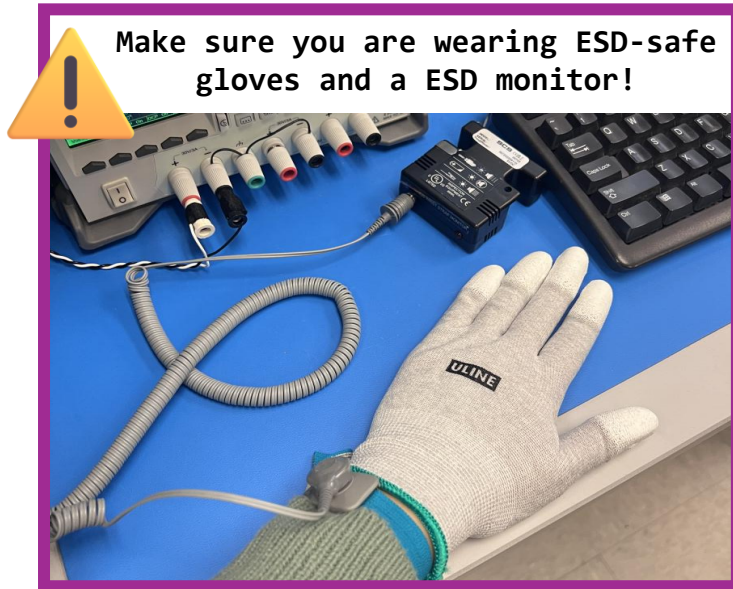
5. Install the Top Power Cable in Slot T



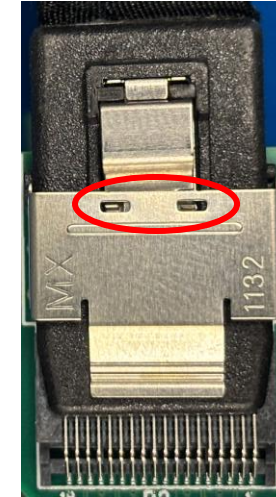
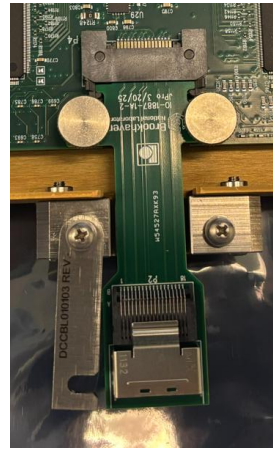
6. Install the Strain Relief Clamp

Top CE Box (HD) Assembly

Follow these steps to assemble the Top CE Box in the support structure:

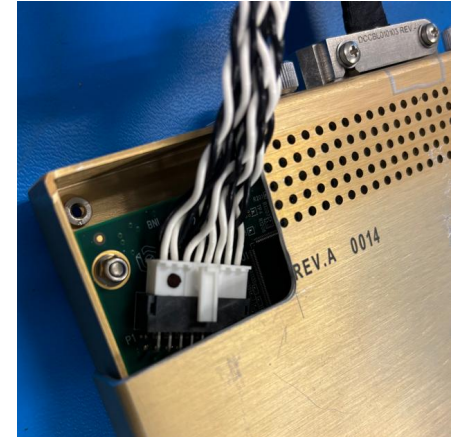


1. Install Adapter + miniSAS Cable

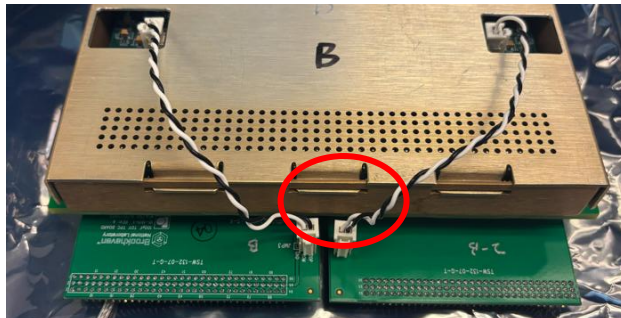


Locks in position

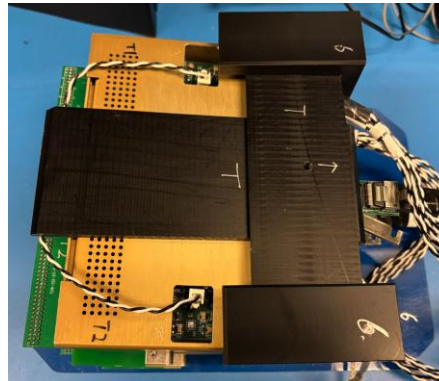
2. Install Test Cover 3. Install Power Cable



3. Remove FE terminators and install Toy TPCs



4. Insert into top Slot and fasten CE box (secure ground strips)

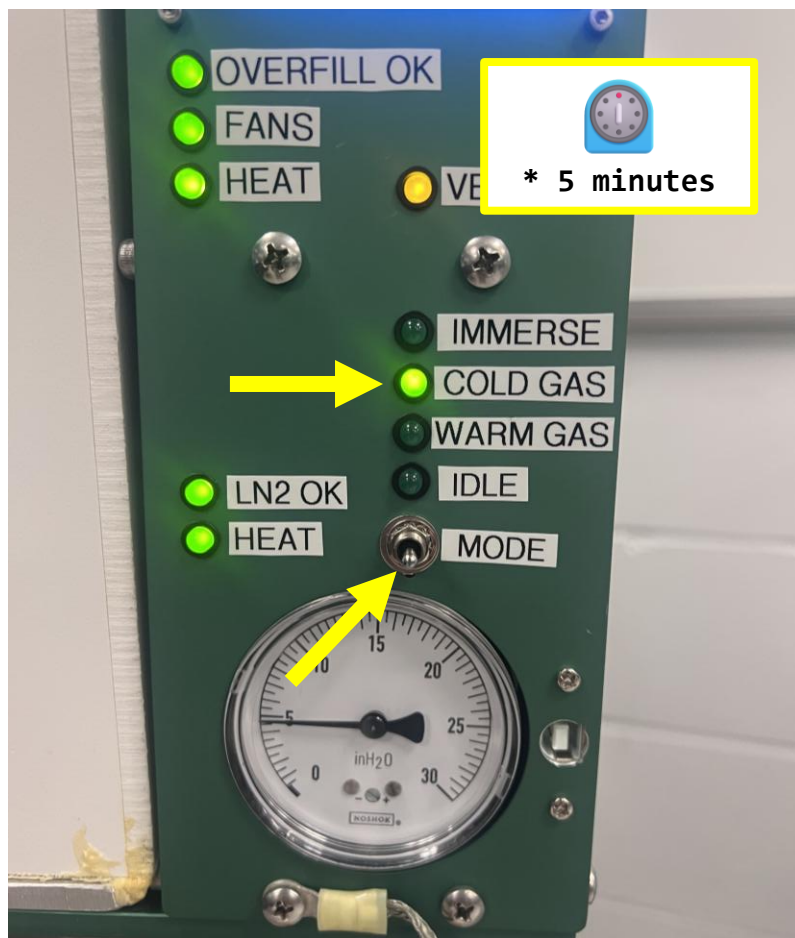


Make sure CTS is closed, locked and ready to use

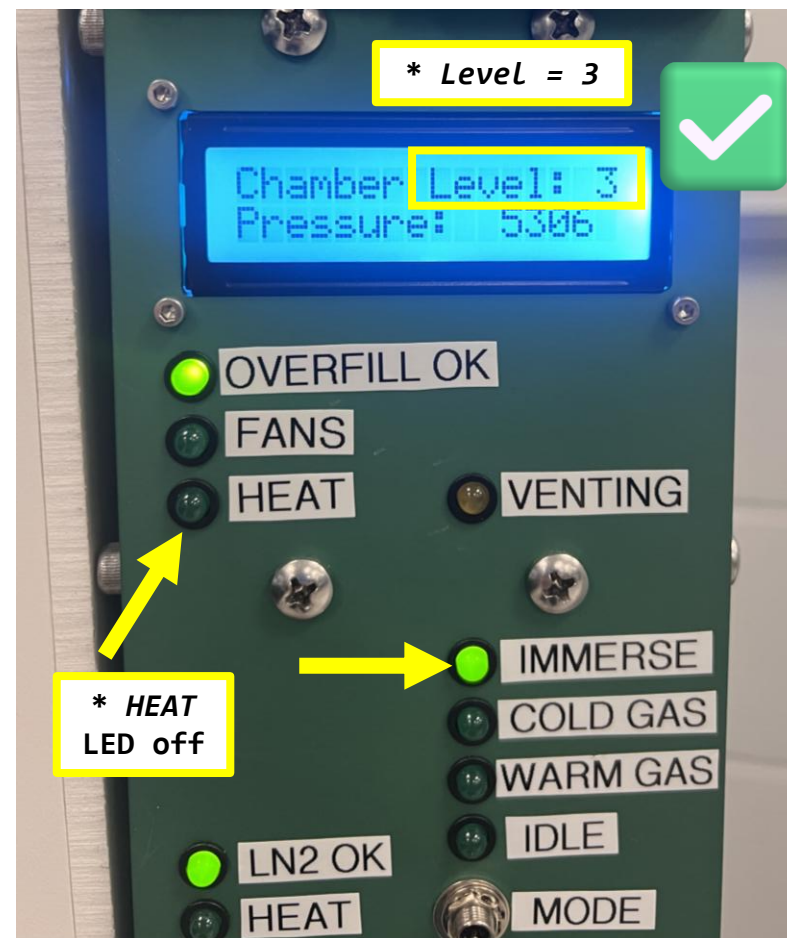


Preparation for Cold QC: CTS Cooldown

1. Switch to *COLD GAS* for 5 minutes

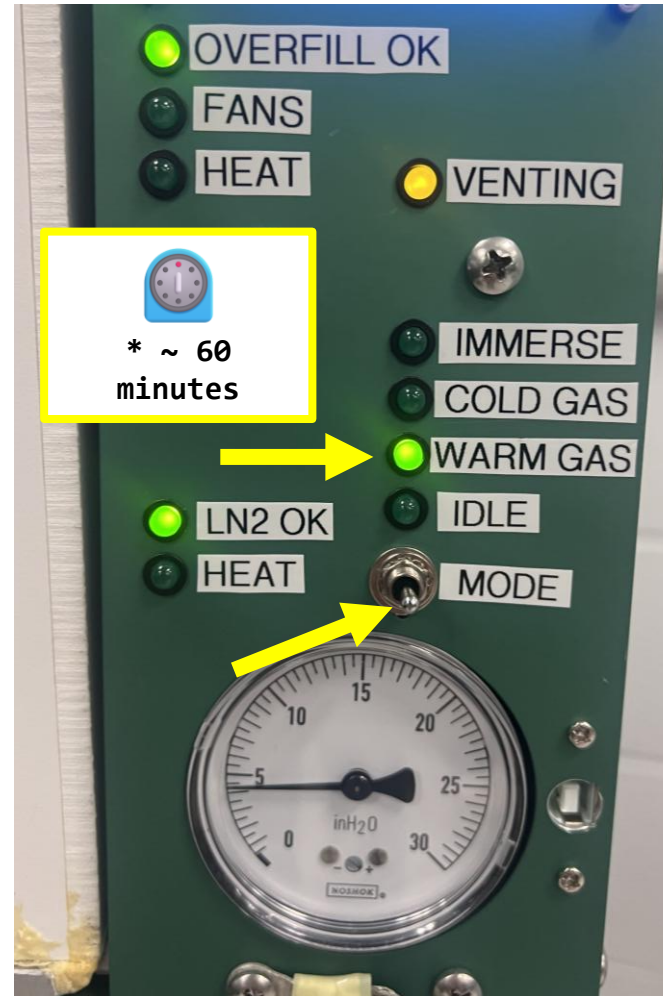


2. Switch to *IMMERSE* and wait for *Level 3* + Heat LED Off



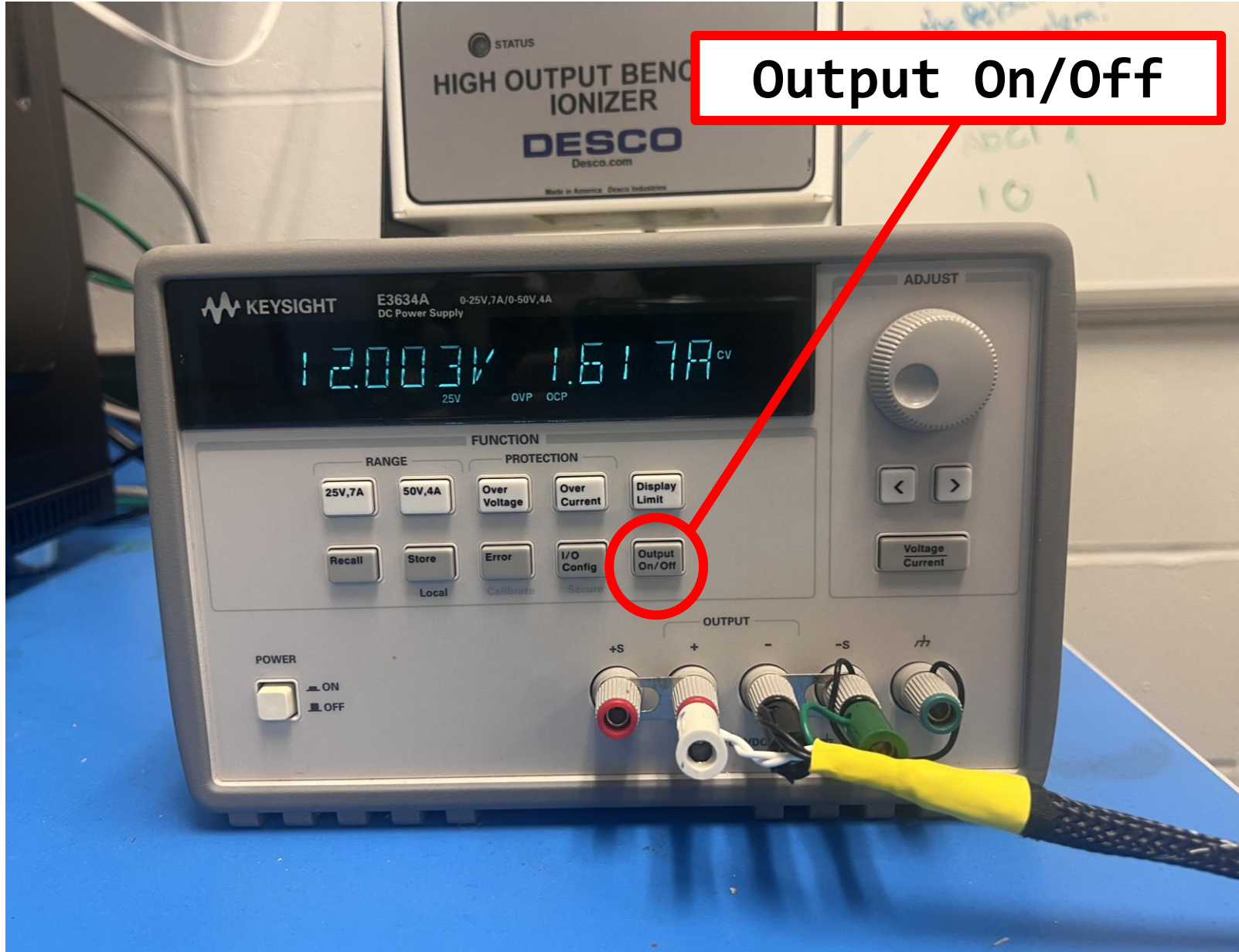
Post-Cold QC: Switch CTS to *WARM*

Switch to *WARM GAS* and
wait 60 minutes

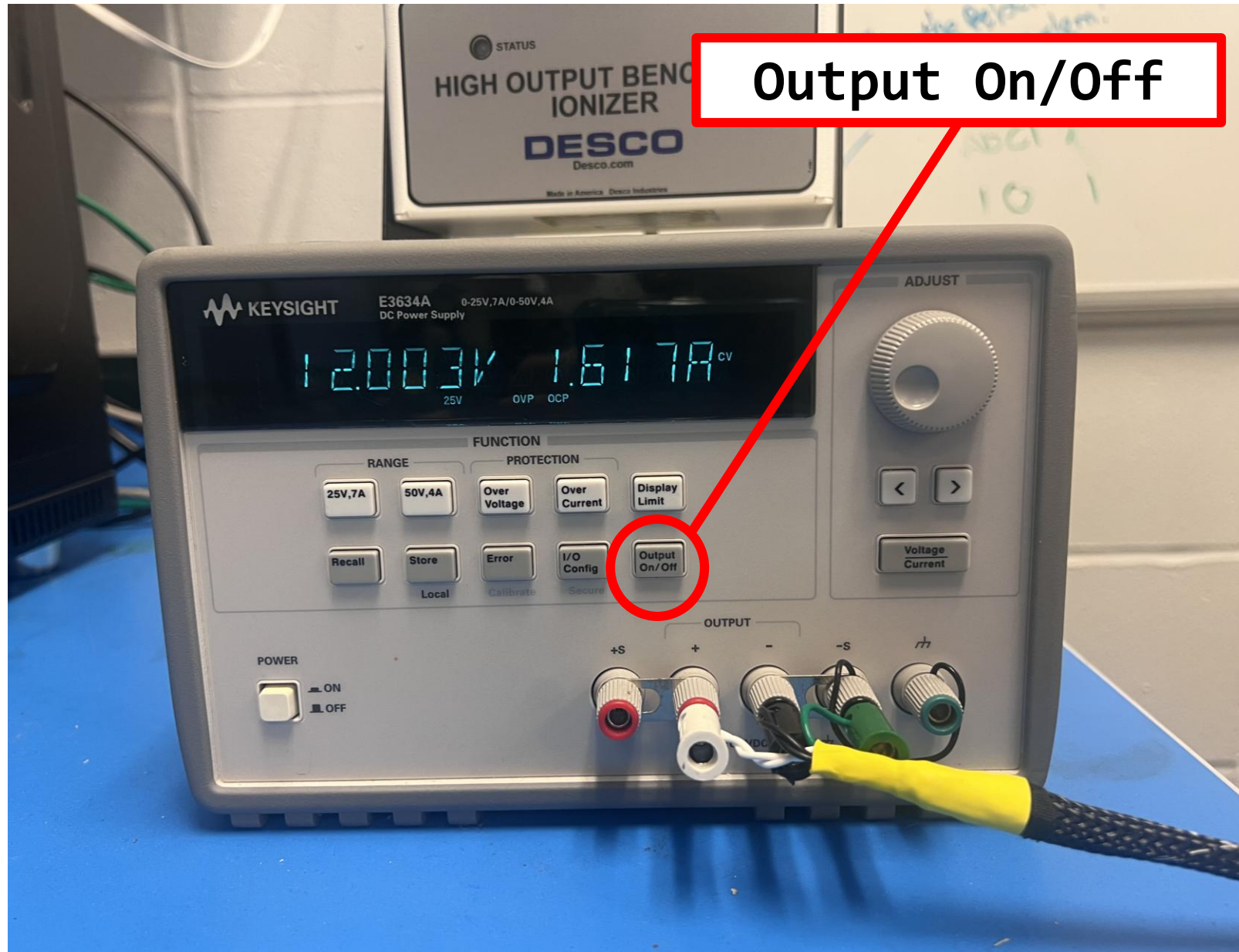


Lingyun: Rigol power supply will be in use, so that computer can control 12V ON/OFF for WIB.
The pop-windows related to power supply should be removed.

Warm QC: Power On WIB



Warm QC is Done! Power Off WIB

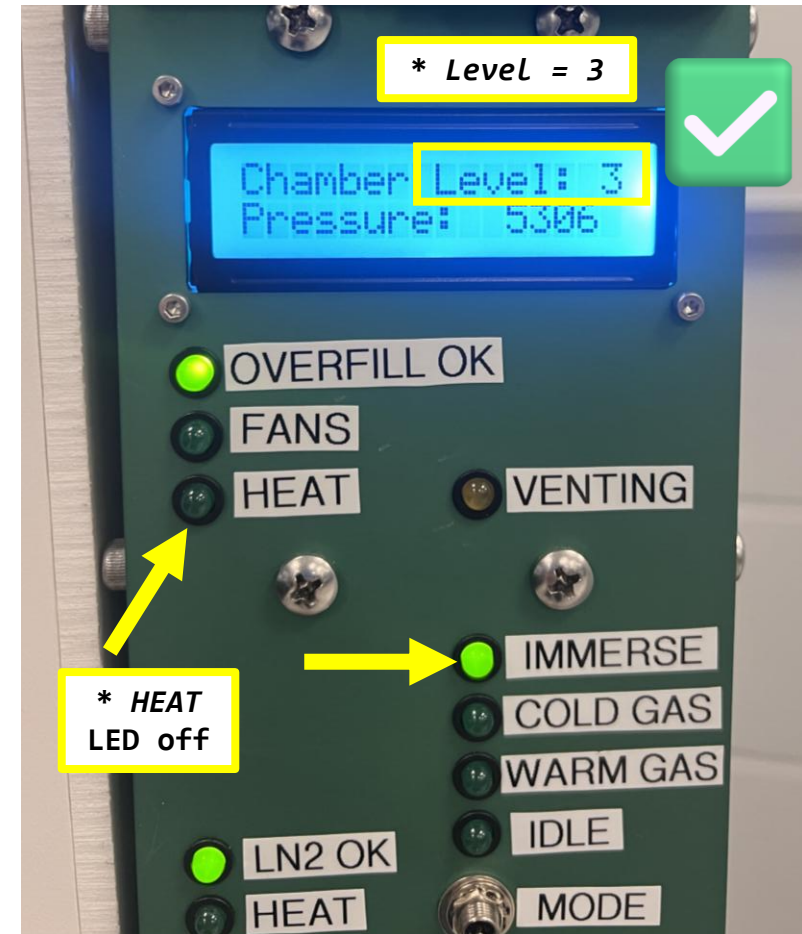


Preparation for Cold QC: CTS Cooldown

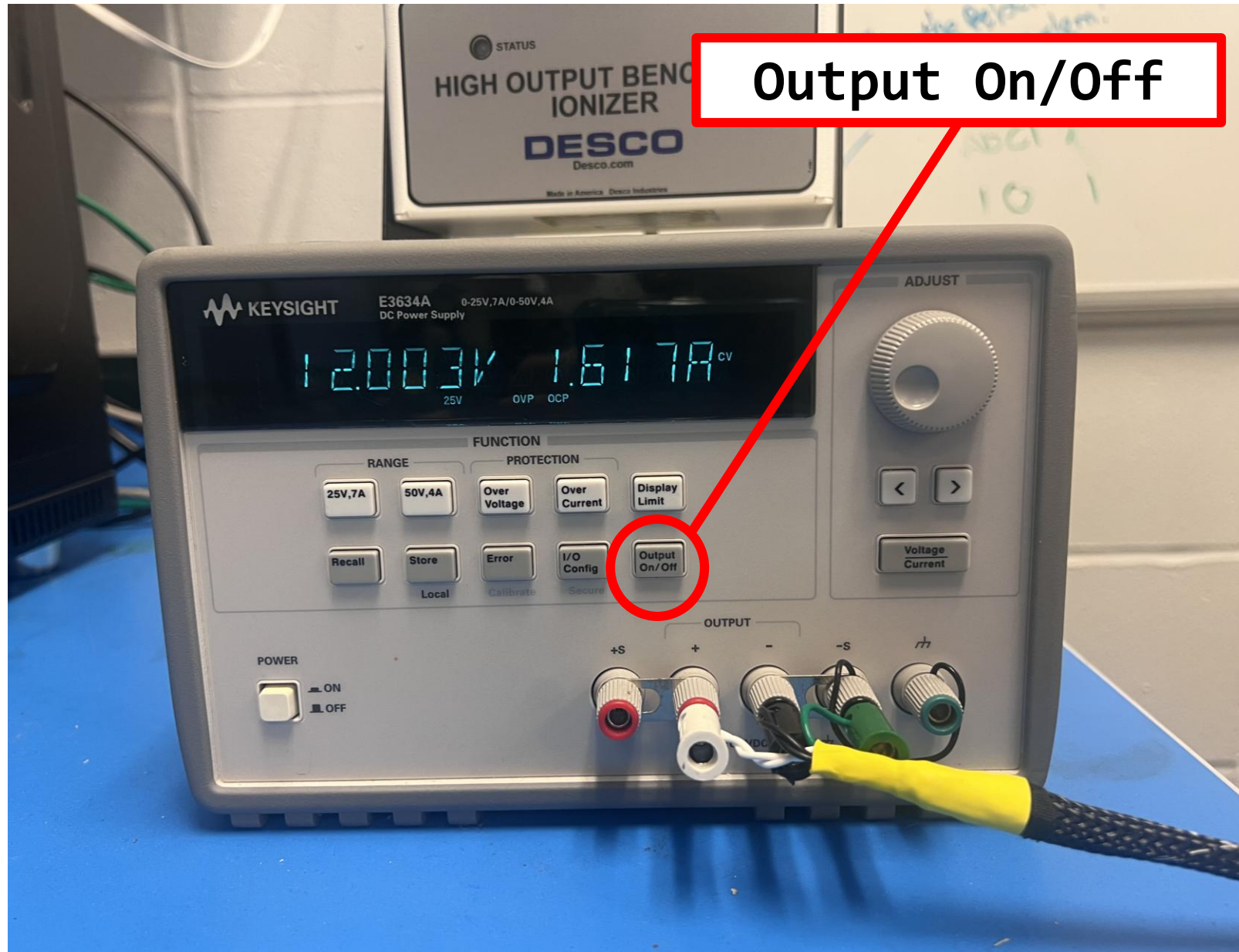
1. Switch to *COLD GAS* for 5 minutes



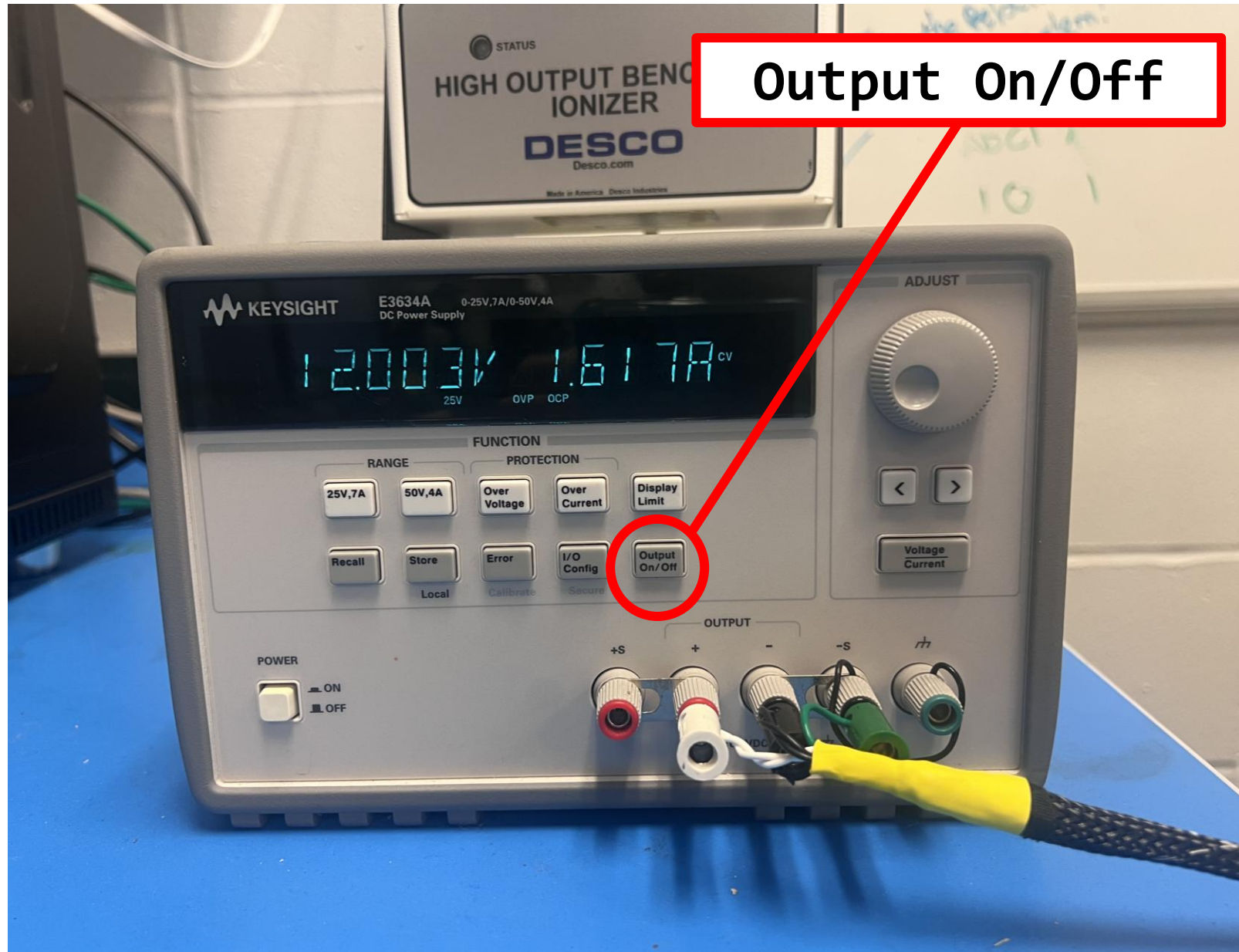
2. Switch to *IMMERSE* and wait for *Level 3* + Heat LED Off



Cold QC: Power On WIB

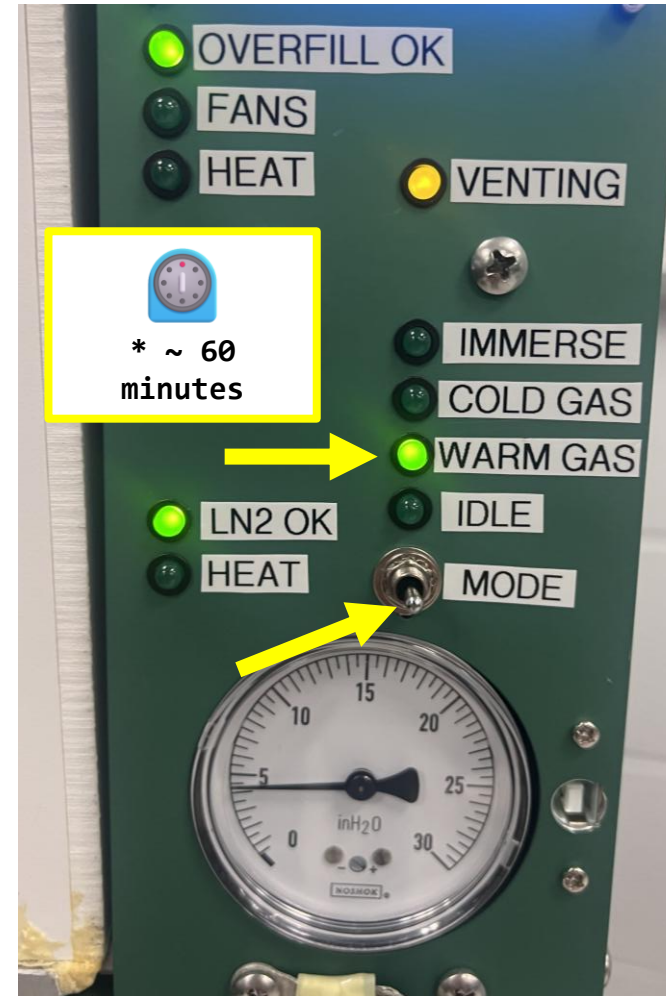


Cold QC is Done! Power Off WIB



Post-Cold QC: Switch CTS to *WARM*

Switch to *WARM GAS* and
wait 60 minutes ~~or~~
~~until the temperature~~
~~reaches 45°C~~



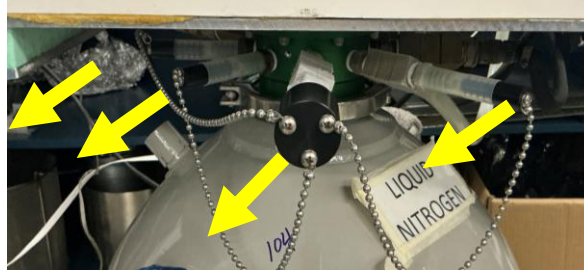
LN2 Filling (**STOP** if you are **NOT** trained)

Follow these steps to fill the CTS dewar with LN2:

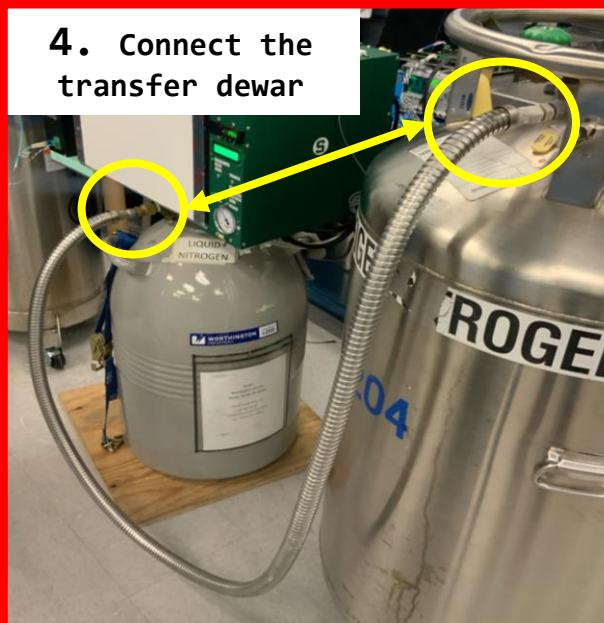
1. Turn On CTS Controller and set in IDLE



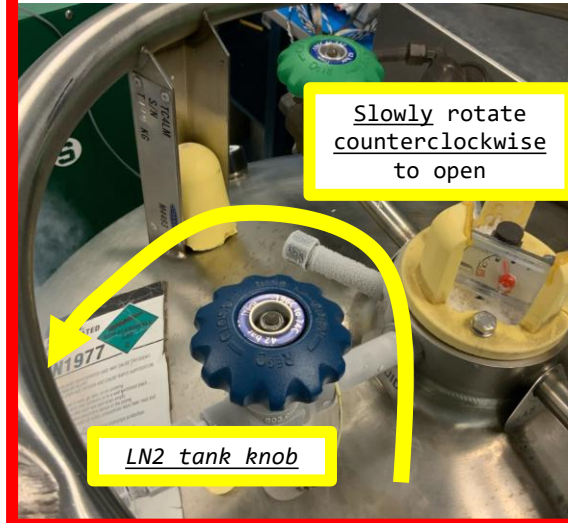
3. Remove the LN2 Dewar plugs
Keep CTS cover close



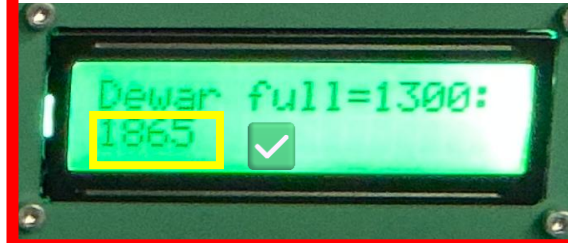
4. Connect the transfer dewar



5. Start the filling process

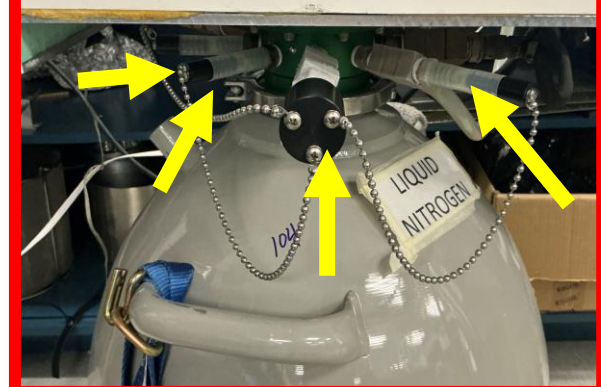


6. Monitor until "Dewar Full"
> 1800

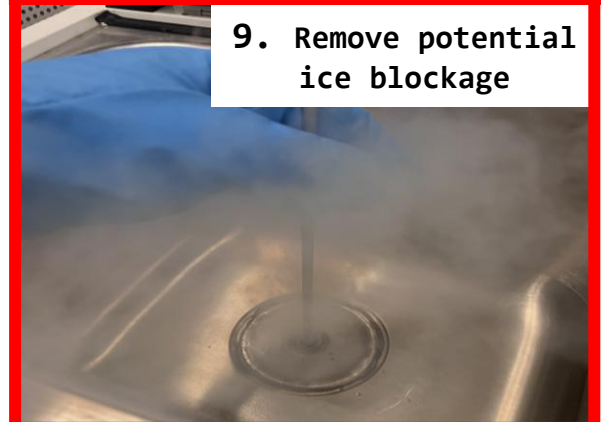


7. Stop the filling process
(rotate LN2 tank knob clockwise)

8. Remove the transfer line and place LN2 Dewar plugs back.

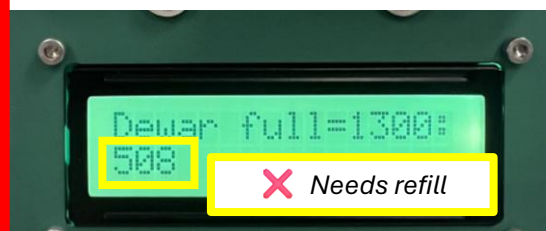


9. Remove potential ice blockage



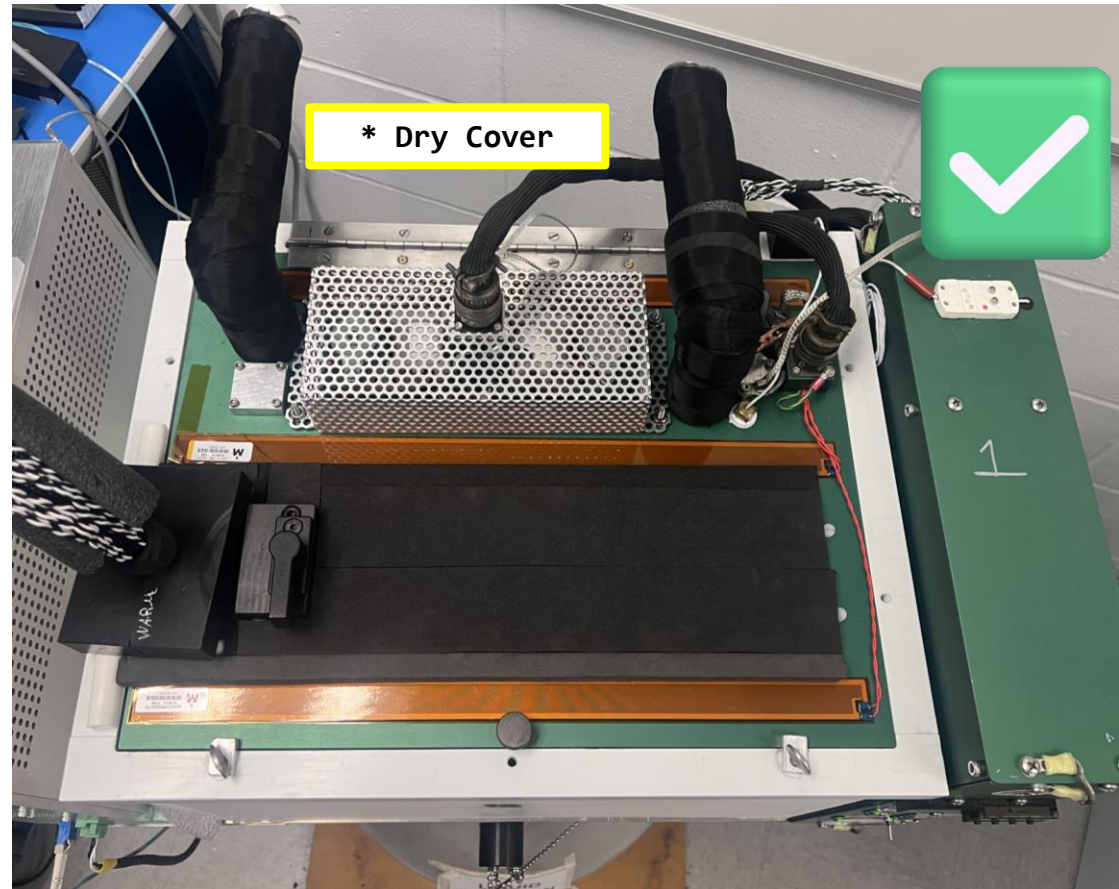
10. Set chamber in warm mode for 15 minutes and then turn CTS OFF.

2. If value for "Dewar Full" is less than 1800, refill. Follow steps 3-10

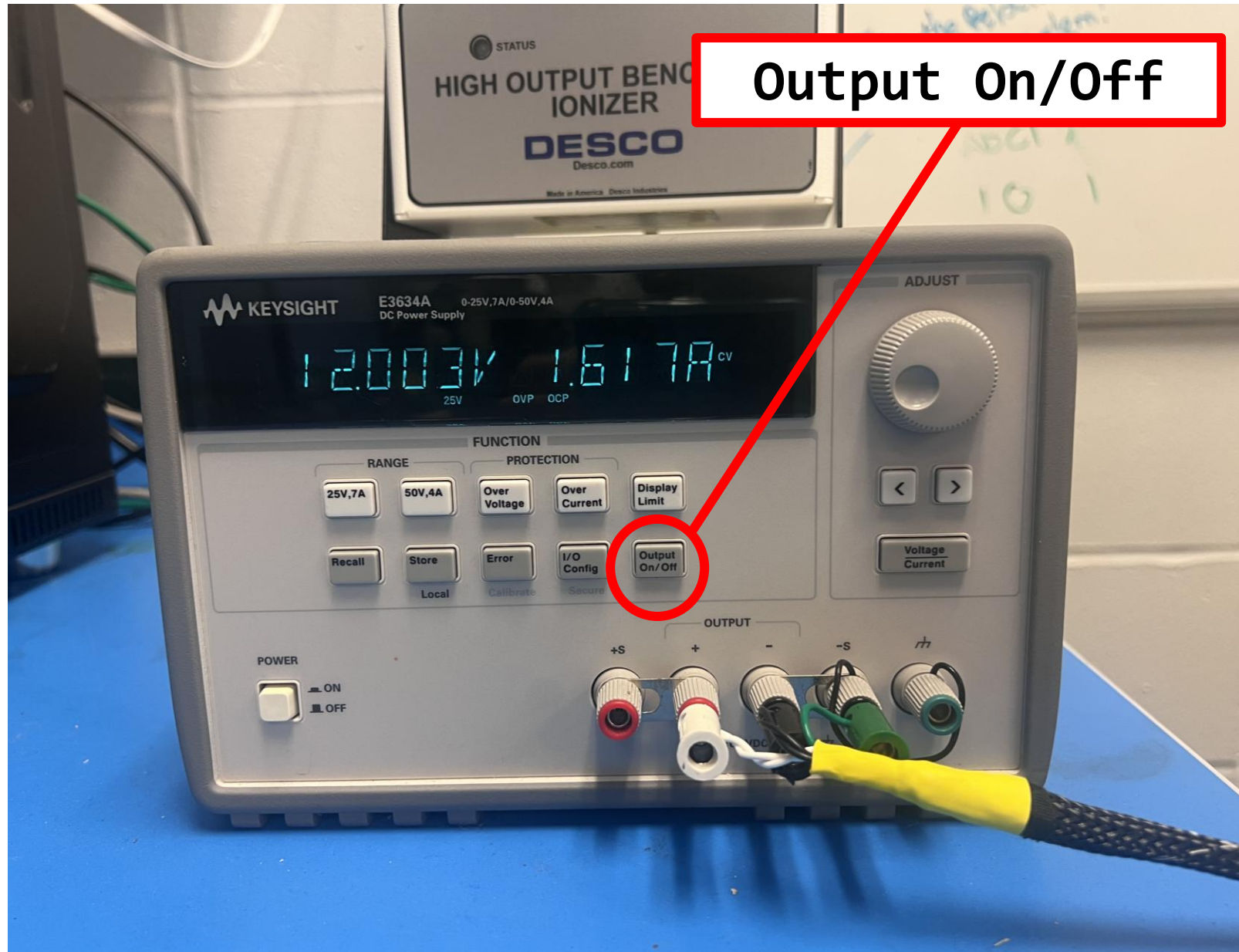


Post-Warm Up Check

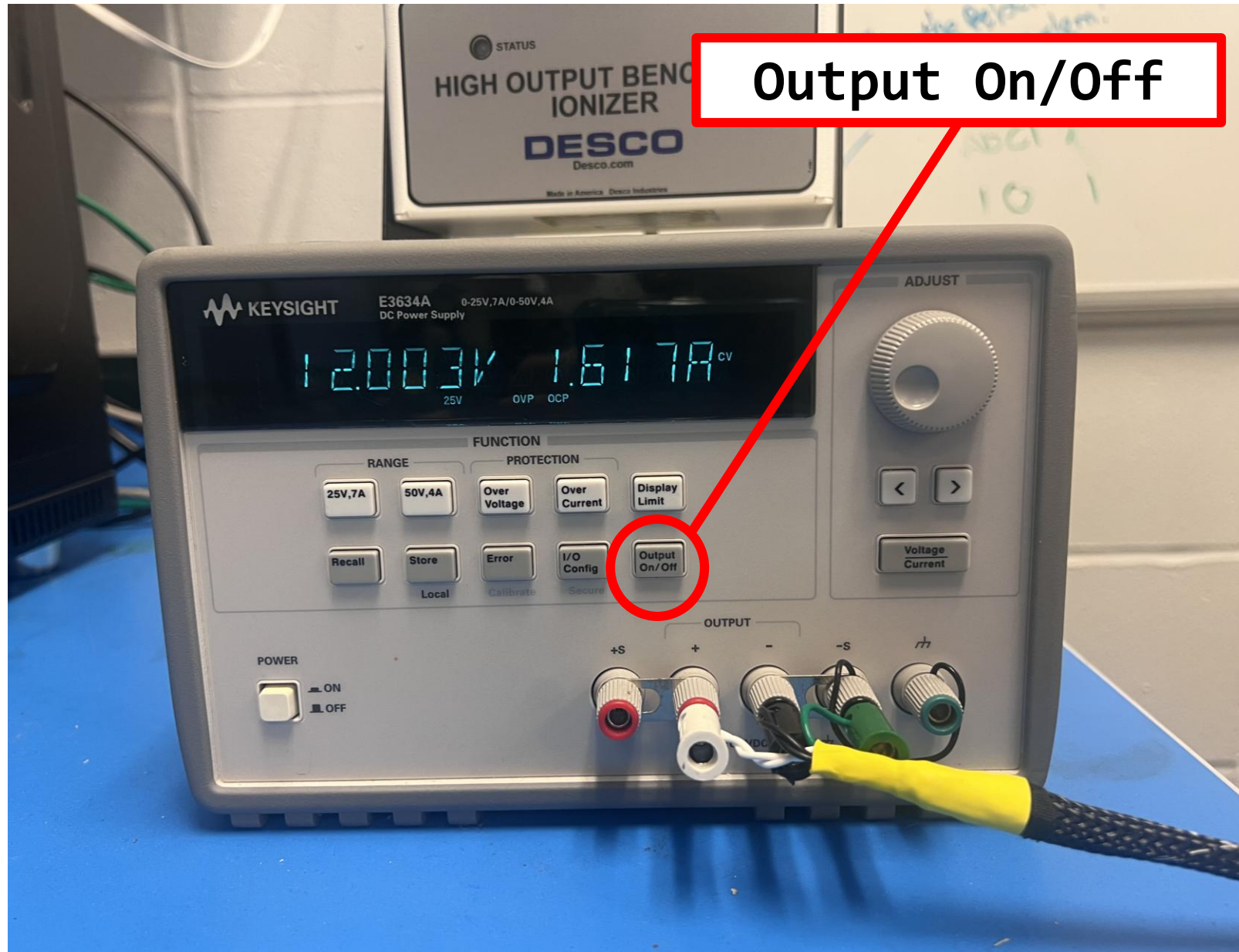
Make sure there is no water on the CTS cover. If there is, wipe it off.



Final Checkout: Turn On WIB

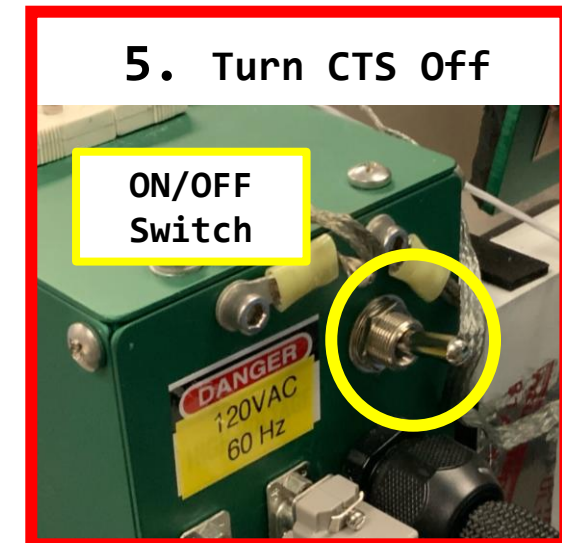
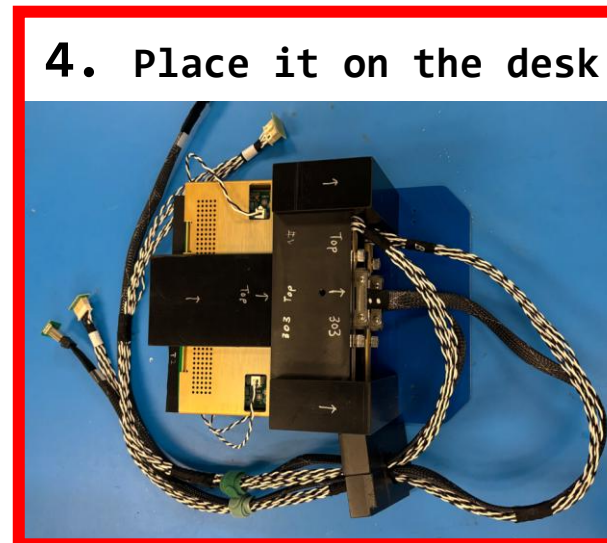
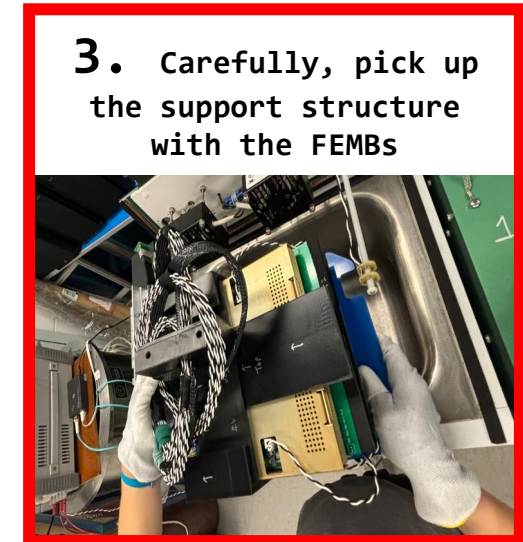
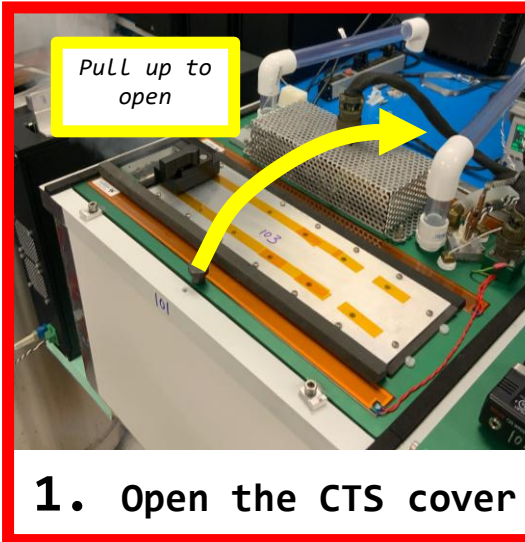
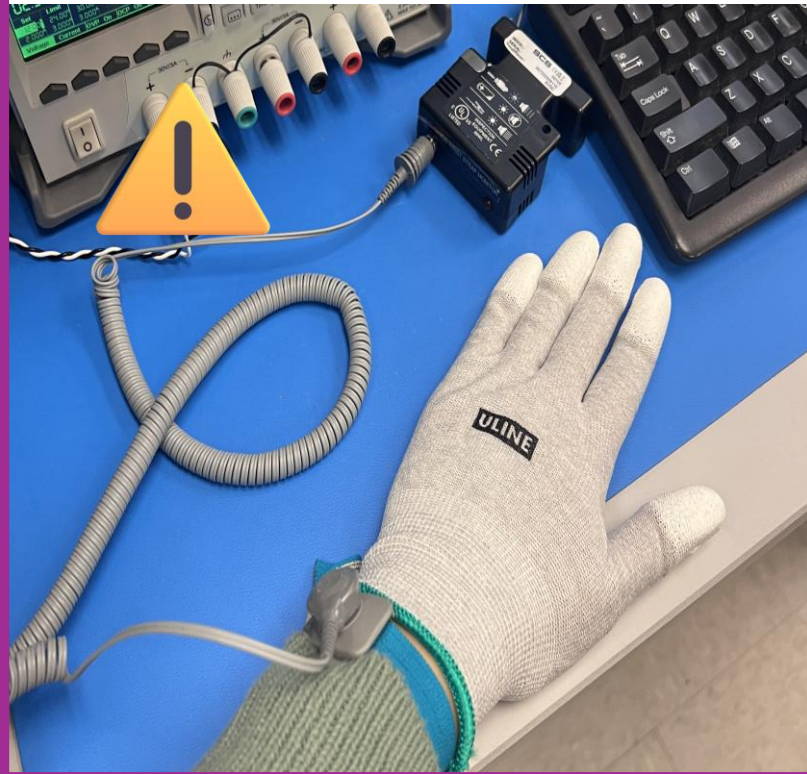


Final Checkout is done! Turn Off WIB



Take the FEMBs out of the CTS Chamber

Remember to wear ESD-safe gloves and a wrist monitor!



Disassembly: Top FEMB

1. Remove the Top CE Box



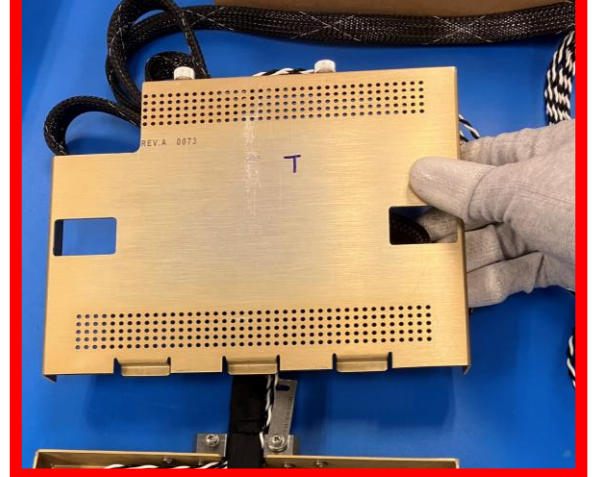
2. Remove Toy TPCs



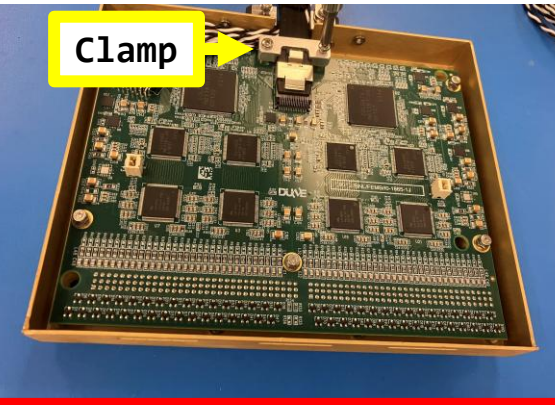
3. Disconnect the Power Cable



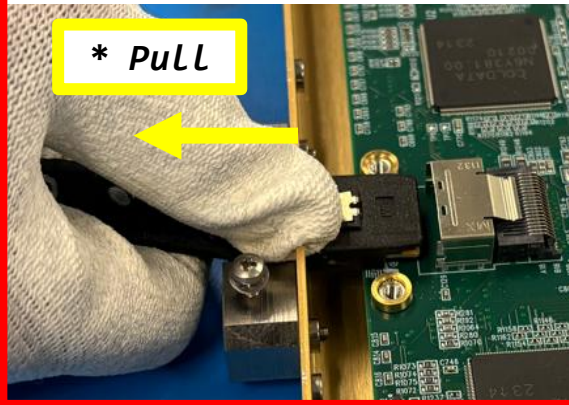
4. Remove the test cover



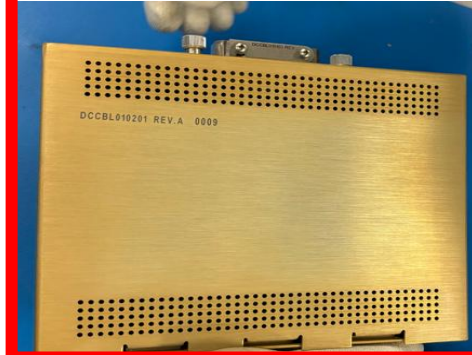
5. Remove the cable clamp



6. Disconnect the Data cable



7. Install the original CE box cover



8. Pack FEMB in ESD Bag and Foam Box

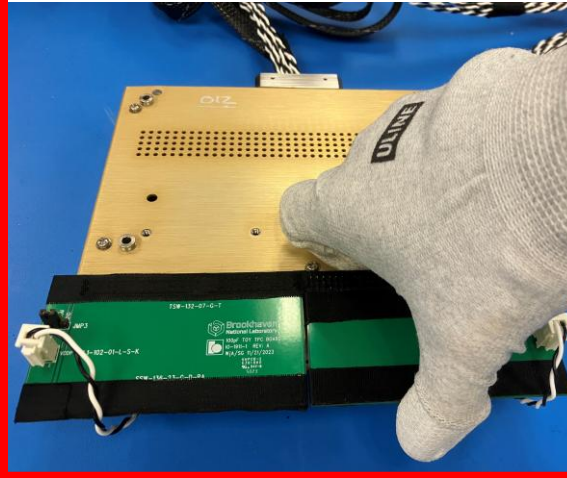


Disassembly: Bottom FEMB

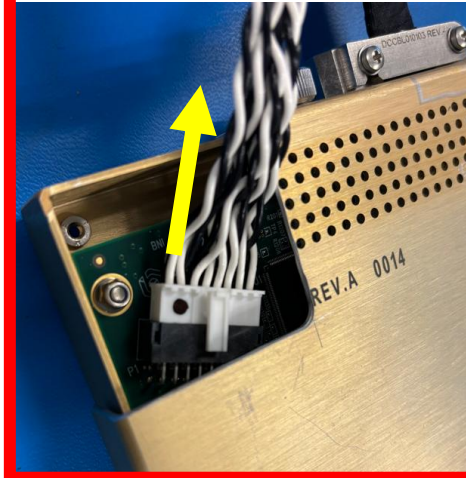
1. Remove the Bottom CE Box



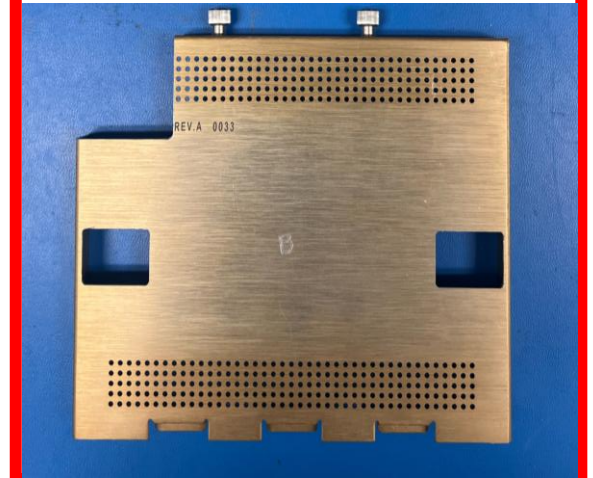
2. Remove Toy TPCs



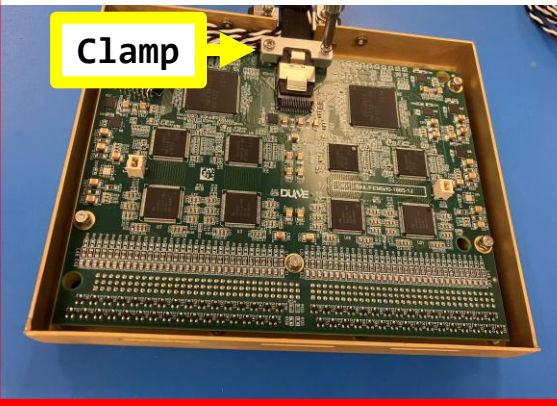
3. Disconnect the Power Cable



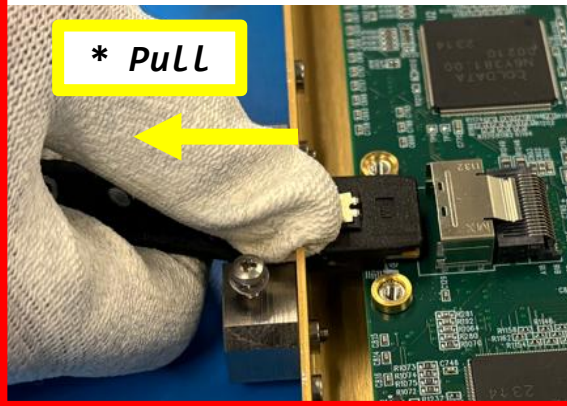
4. Remove the test cover



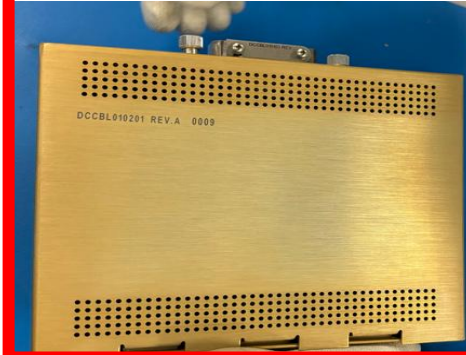
5. Remove the cable clamp



6. Disconnect the Data cable



7. Install the original CE box cover



8. Pack FEMB in ESD Bag and Foam Box



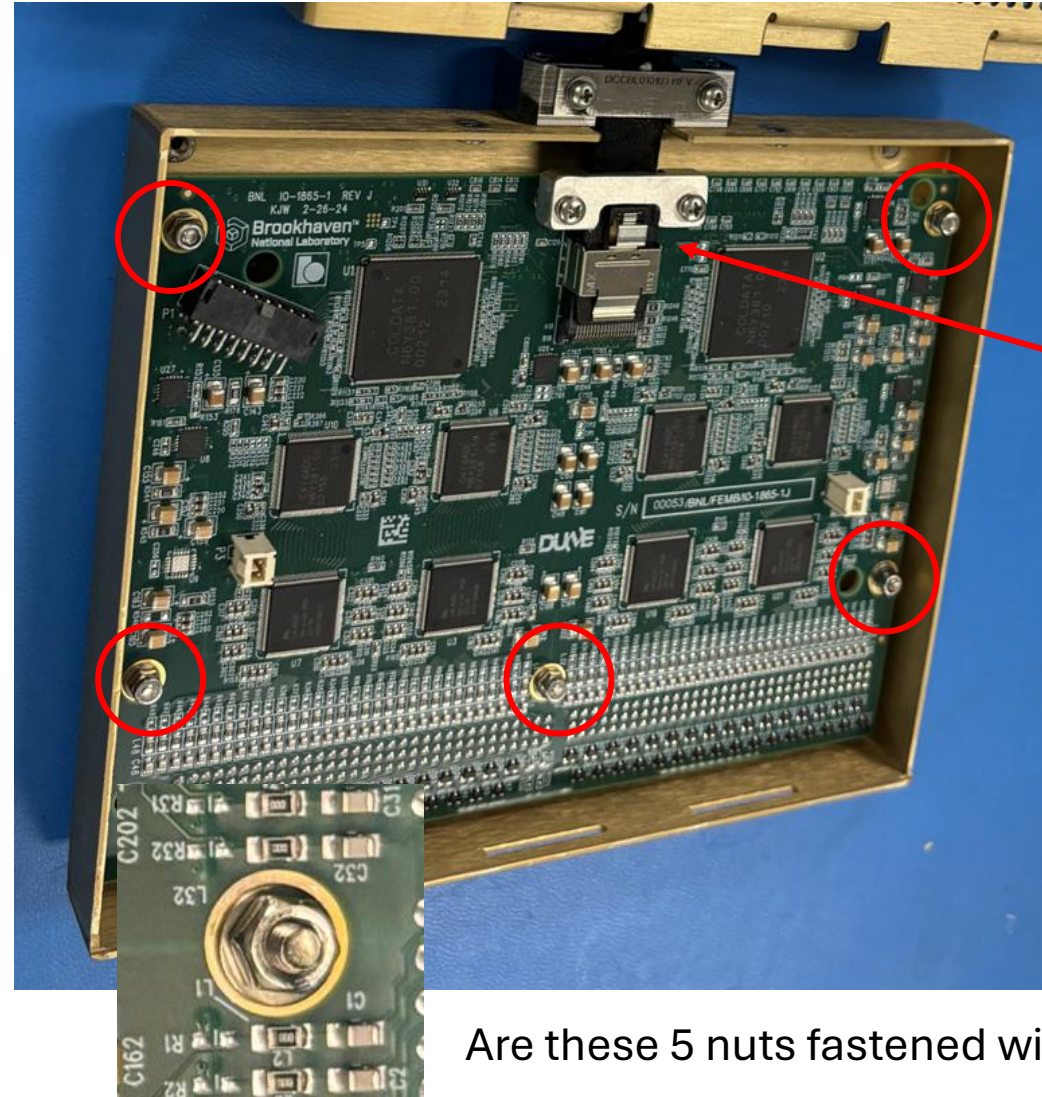
CE Box Visual Inspection



If foam box was broken?
If HWDB QR sticker was missing?



CE box enclosed by ESD bag?



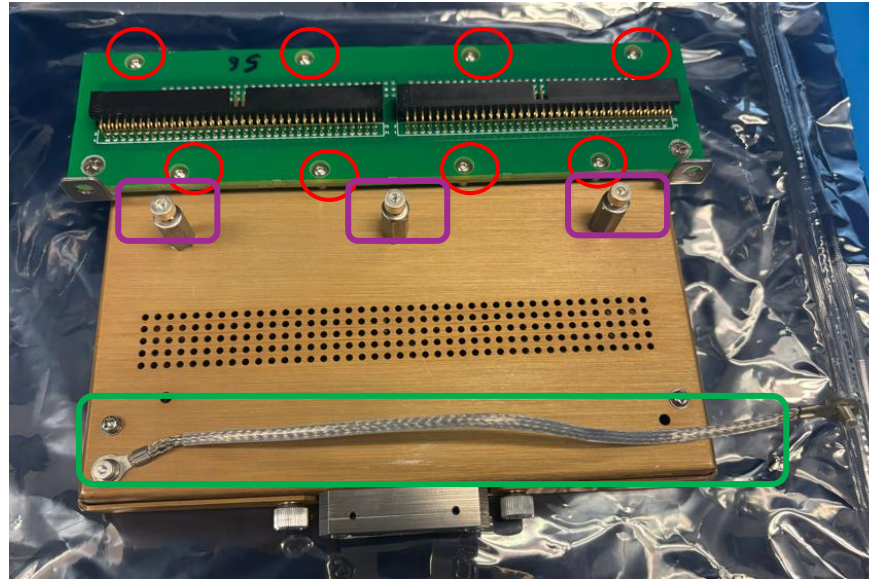
Connector
clamp in position

Are these 5 nuts fastened with washers?

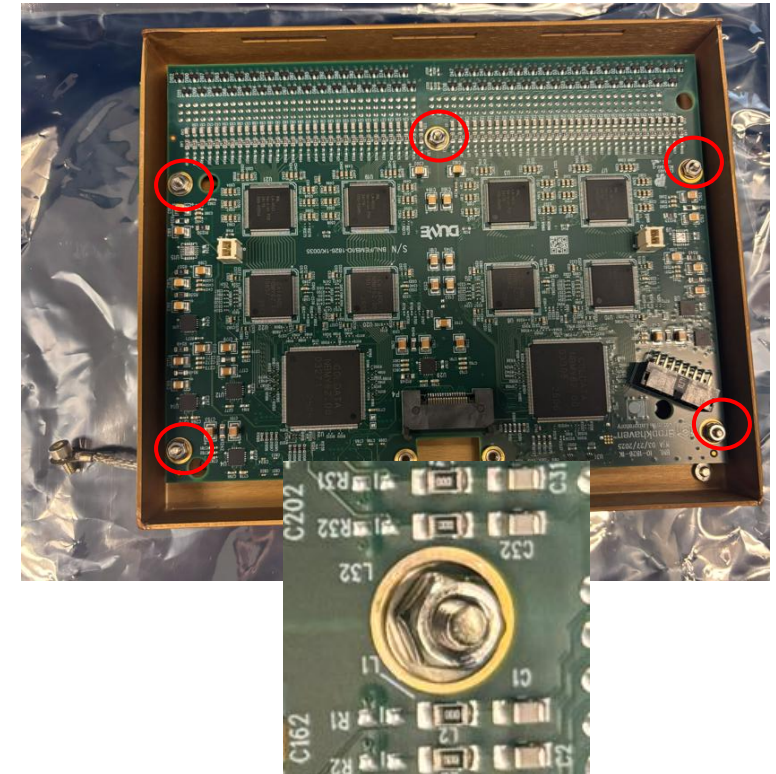
Bottom CE Box (HD) Visual Inspection



If foam box was broken?
If HWDB QR sticker was missing



Nuts for CR-RC adapter board in position?
3 sets of lock nut and washer in position?
Ground strip in position?
FE input terminator in position



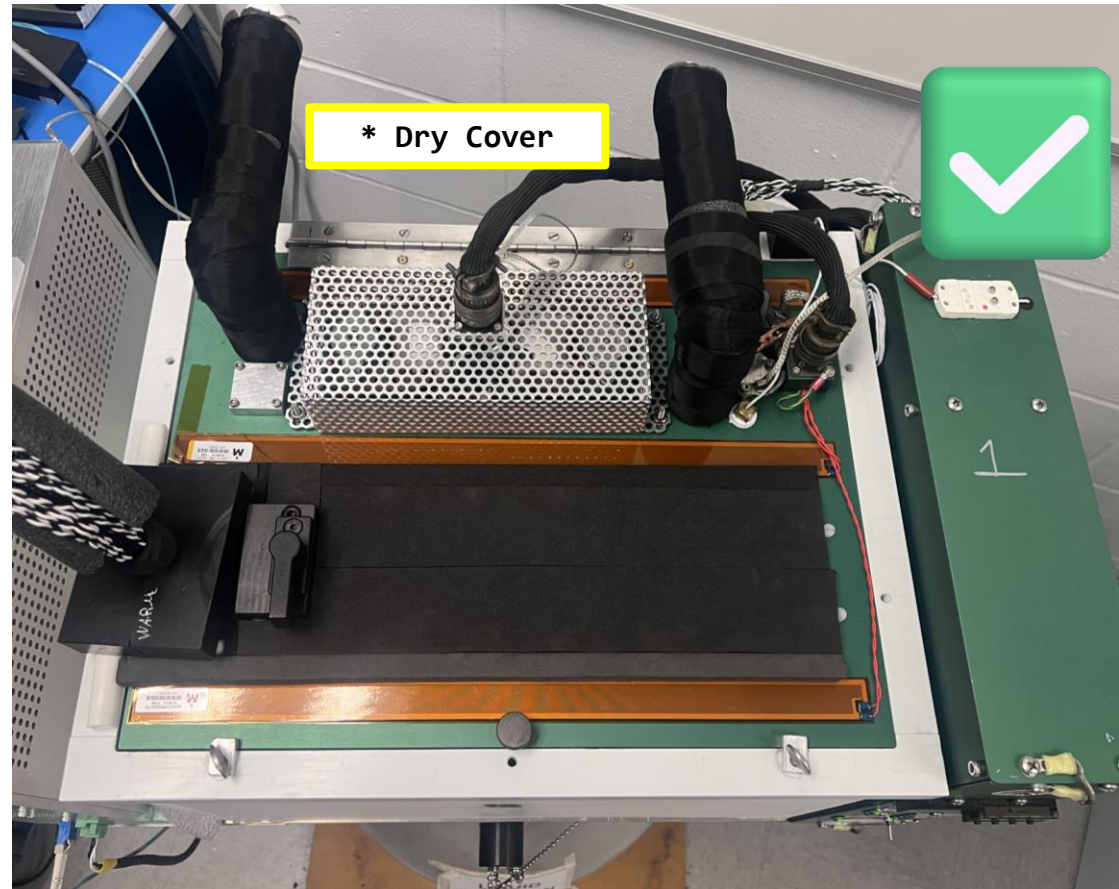
Are these 5 nuts fastened with washers?



CE box enclosed by ESD bag?

Post-Warm Up Check

Make sure there is no water on the CTS cover. If there is, wipe it off.



Disassembly: Bottom CE box (HD)

1. Remove the Bottom CE Box



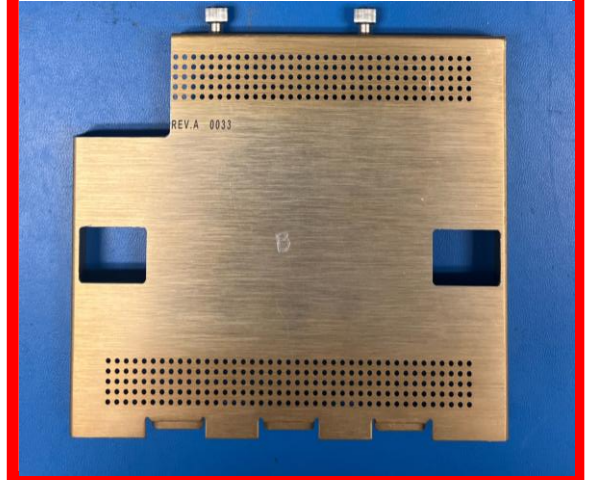
2. Remove Toy TPCs



3. Disconnect the Power Cable



4. Remove the test cover



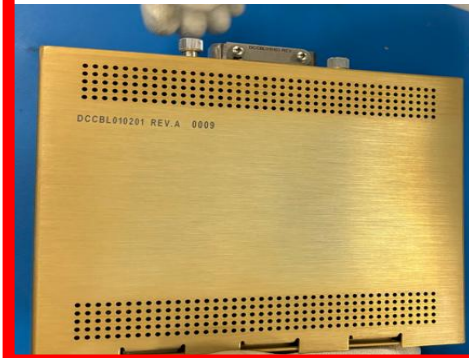
5. Remove the miniSAS cable



6. Place the connector clamp back on FEMB if it comes with CE box



7. Install the original CE box cover



8. Pack FEMB in ESD Bag and Foam Box

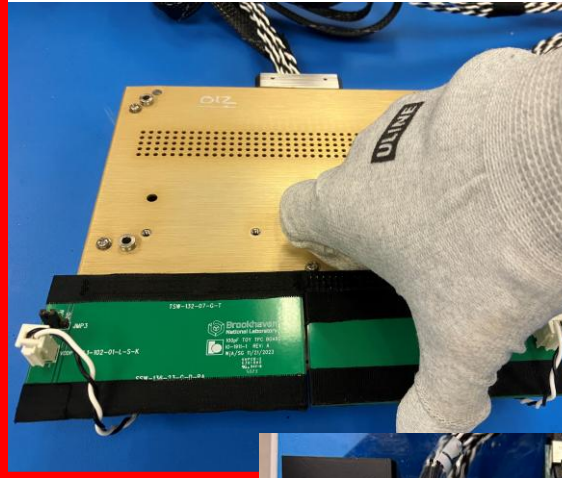


Disassembly: Top CE box (VD)

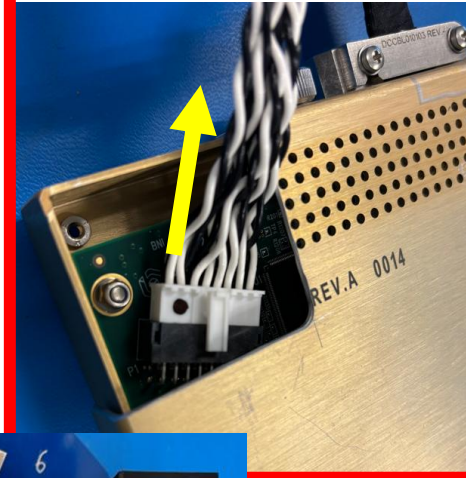
1. Remove the Top CE Box



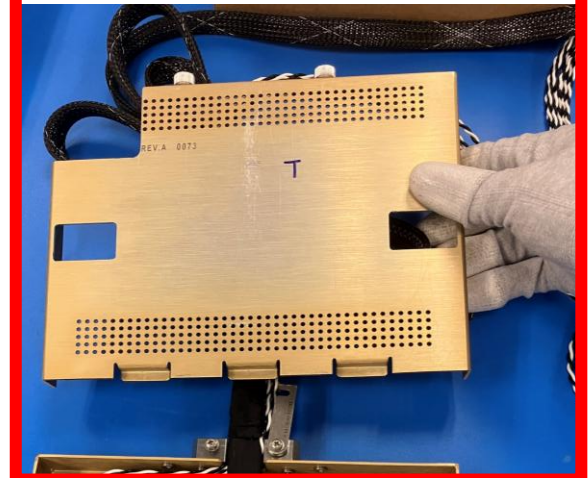
2. Remove Toy TPCs



3. Disconnect the Power Cable



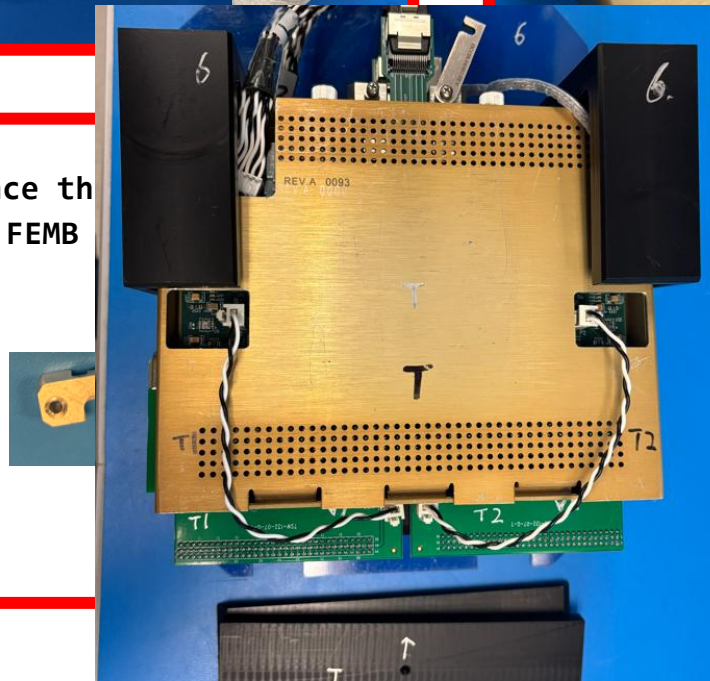
4. Remove the test cover



5. Remove the miniSAS cable



6. Place the back on FEMB CE box



7. Install the box cover



8. Pack FEMB in ESD Bag and Foam Box



Disassembly: Bottom CE box (VD)

1. Remove the Bottom CE Box



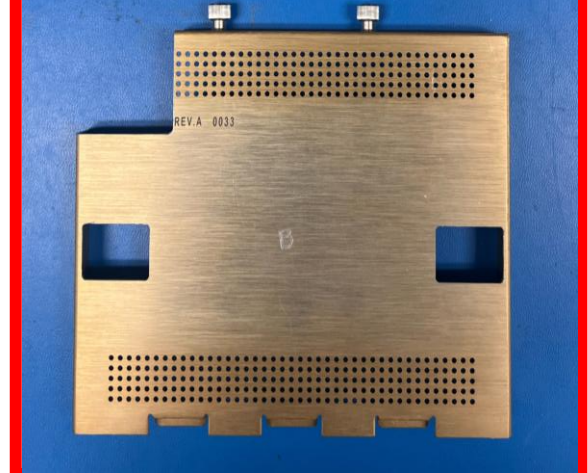
2. Remove Toy TPCs



3. Disconnect the Power Cable



4. Remove the test cover



5. Remove the miniSAS cable



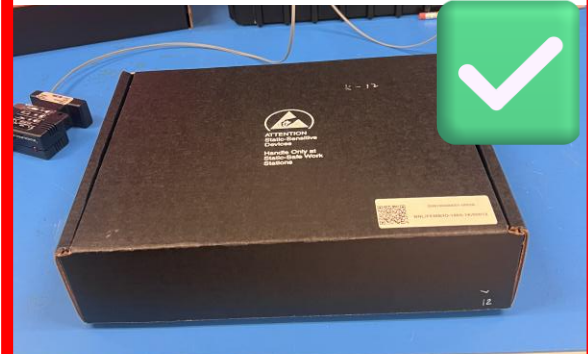
6. Place the connector clamp back on FEMB if it comes with CE box



7. Install the original CE box cover



8. Pack FEMB in ESD Bag and Foam Box



CE Box Visual Inspection

Please contact Tech Coordinator if abnormality is observed!

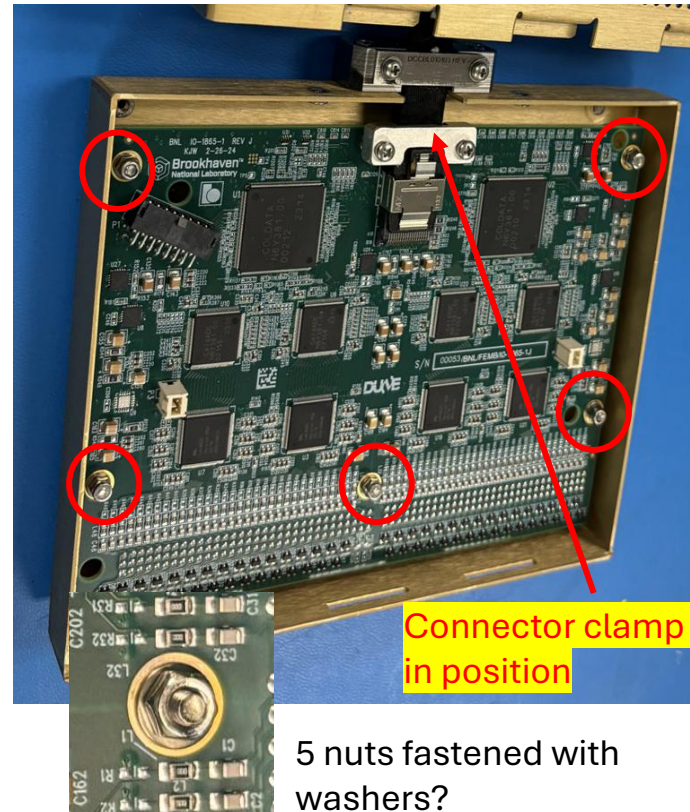


If foam box was broken?
If HWDB QR sticker was missing?



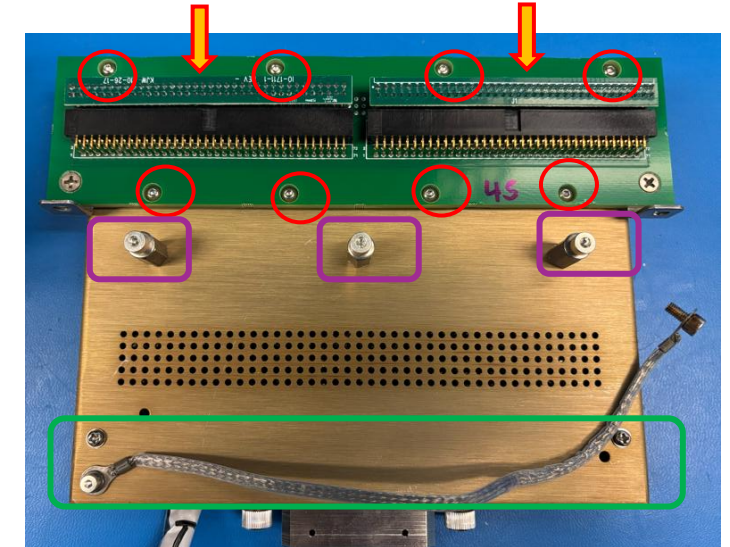
CE box enclosed by ESD bag?

VD CE box (1865)

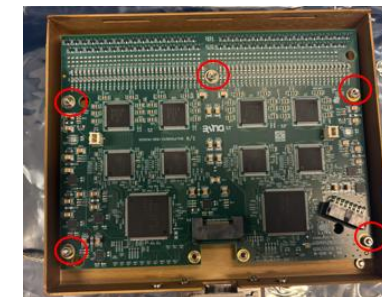


5 nuts fastened with washers?

HD CE box (1826)



6 Nuts for CR-RC adapter board?
3 sets of lock nut and washer?
1 Ground strip?
2 FE input terminators?



5 nuts fastened with washers?